BRW via Analytic Combinatorics: Complete Worked Example

From the Chemical Equation to Critical Exponents
with Full Proofs and Numerical Verification

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Abstract

We present a self-contained, fully worked derivation of the Branching Random Walk (BRW) scaling exponents using analytic combinatorics (AC). Starting from the chemical equations, we construct the Liouvillian, extract vertices, build the Dyson–Schwinger equation as a Lagrange equation, locate the branch point, apply the transfer theorem, invoke Weyl's law, and arrive at the critical exponents $\alpha_p = (pd_s - 2)/2$. Every algebraic step is shown explicitly. We verify numerically against simulation data on $\mathbb{Z}^d$ lattices, Sierpinski carpets, random trees, and scale-free networks. Symmetry factors are computed for representative diagrams. No Feynman diagram loop integral is evaluated at any point: the AC route replaces momentum-space integration with singularity analysis of the generating function. We are careful to distinguish which steps are rigorous theorems (Lagrange inversion, transfer theorem, Weyl's law), which are established results from probability theory (critical branching survival, Dvoretzky–Erdős volume growth), and which are physically motivated connections validated by simulation (the moment factorisation, the spectral dimension substitution).

Contents

1 Overview: What We Will Do and Why

1.1 The problem

A **branching random walk** (BRW) is a cloud of particles that hop on a graph, reproduce, and die. A natural question: *how does the number of distinct sites visited scale with time?* The answer is a **critical exponent** α_p defined by $\langle a^p \rangle(t) \sim t^{\alpha_p}$, where a is the number of visited sites and p is the moment order.

Computing α_p by the standard field-theoretic route requires evaluating Feynman diagram loop integrals and performing renormalisation group (RG) analysis. We show that the same exponents can be obtained *without evaluating any loop integrals*, using analytic combinatorics.

1.2 What is analytic combinatorics?

Analytic combinatorics (AC) [6] is a method for counting combinatorial objects (trees, graphs, permutations, ...) by:

1. Encoding the objects as a **generating function** (a formal power series $A(z) = \sum_n a_n z^n$ where a_n counts objects of size n).
2. Finding the **singularities** of $A(z)$ in the complex plane (the points where $A(z)$ ceases to be analytic).
3. Using the **transfer theorem** to read off the asymptotic growth $a_n \sim C\, n^\theta \rho^{-n}$ directly from the singularity type, without summing the series.

The key insight: *the singularity type of the generating function determines the asymptotic growth of its coefficients.* A square-root branch point $\sqrt{1 - z/z^*}$ always gives $a_n \sim n^{-3/2}$; a simple pole $1/(1 - z/z^*)$ always gives $a_n \sim$ const. The prefactors depend on the specific problem, but the exponent depends only on the singularity type.

For readers unfamiliar with AC, Appendix A provides a self-contained tutorial with worked examples.

1.3 The pipeline

Our derivation follows a 15-step chain. Here is the roadmap, so the reader knows where each section is heading:

Step	What we compute	Method	Section
1–2	Chemical equations → stoichiometry	Input	2
3–4	Stoichiometry → vertices	Doi formula (algebra)	3–4
5	Vertices → DSE kernel $\phi(G)$	Lagrange equation	5
6	Branch point of $\phi(G)$	Implicit function thm	6
7	Coefficient asymptotics $n^{-3/2}$	Transfer theorem	7
8	Survival probability t^{-1}	Branching process theory	8
9–11	Diffusion volume $t^{d/2}$	Weyl's law + Laplace	9
—	Two-species structure $(A + B)$	Why A-sector suffices	10
12–15	Exponents $\alpha_p = (pd-2)/2$	Combine + validate	11

Steps 1–7 are the “AC half” (pure algebra and singularity analysis). Steps 8–11 are the “spectral half” (branching process theory and spectral geometry). Step 12 combines them. Steps 13–15 validate against simulation.

2 The Chemical Equation

The BRW consists of two species: **active walkers** A that hop on a graph with diffusion constant D, and **immobile tracers** B deposited by walkers with site exclusion (at most one B per site). The observable a (number of distinct sites visited) is the total number of B particles. The tracers are passive: they do not affect the A dynamics.

The full BRW has **four** reactions — three in the A-sector (the Gribov process) and one transmutation coupling A to B:

$$\boxed{\begin{array}{ll} A \xrightarrow{\beta} 2A & \text{(branching, rate } \beta) \\ A \xrightarrow{\epsilon} \emptyset & \text{(death, rate } \epsilon) \\ 2A \xrightarrow{\chi} A & \text{(coagulation, rate } \chi) \\ A \xrightarrow{\Lambda} A+B & \text{(transmutation, rate } \Lambda\text{, with site exclusion)} \end{array}} \tag{1}$$

The first three reactions form the **Gribov process** (the A-sector). The fourth reaction deposits a B tracer at the walker's site, but only if no B is already present there (site exclusion: $n_B \leq 1$). The site exclusion makes this a *conditional* reaction, encoded by the carrying capacity trick (Section 10).

The A-sector stoichiometry matrix is:

$$S_A = \begin{pmatrix} 1 & 2 \\ 1 & 0 \\ 2 & 1 \end{pmatrix} \tag{2}$$

where row i is (k_i, ℓ_i) = (reactants, products) for reaction i.

Remark. The A-sector stoichiometry S_A determines the Gribov vertices (Sections 3–4). The transmutation reaction produces 6 additional multi-species vertices (Section 10), but these do not modify the A-sector singularity (Proposition 10.1). We treat the A-sector first and return to the full two-species structure in Section 10.

Remark. The A-sector stoichiometry matrix is the *only* input to the AC pipeline for computing exponents. Everything that follows — the Liouvillian, the vertices, the DSE, the branch point, the exponents — is derived algorithmically from S.

3 The Liouvillian

3.1 The Heisenberg–Weyl Algebra

Definition 3.1 (Heisenberg–Weyl algebra)**.** The Heisenberg–Weyl algebra $\mathfrak{h}$ is generated by ∂_z and z with the Lie bracket

$$[\partial_z, z] = 1 \qquad \text{i.e.} \qquad \partial_z \cdot z - z \cdot \partial_z = 1 \tag{3}$$

In the Doi picture [1]: $z \leftrightarrow a^\dagger$ (creation/multiplication) and $\partial_z \leftrightarrow a$ (annihilation/differentiation).[1]

[1] The identification is: z acts by multiplication (adding a particle to the state, like $a^\dagger$), while ∂_z acts by differentiation (removing a particle, like a). The commutation relation $[\partial_z, z] = 1$ is the Leibniz rule $\partial_z(zf) = f + z\partial_z f$, which is algebraically identical to $[a, a^\dagger] = 1$.

3.2 The Doi–Peliti Generator Formula

Theorem 3.2 (Single-reaction generator [1, 2, 4]). *For a reaction $kA \to \ell A$ at rate λ, the infinitesimal generator of the master equation in the factorial-moment representation is:*

$$\boxed{Q[\partial_z, z] = \lambda\big((z+1)^\ell - (z+1)^k\big)\partial_z^k} \tag{4}$$

Proof. The master equation for the occupation number probability $P(n,t)$ is

$$\partial_t P(n,t) = \lambda\binom{n+k-\ell}{k}P(n+k-\ell,t) - \lambda\binom{n}{k}P(n,t)$$

The factorial-moment generating function $\Psi(z,t) = \sum_n (1+z)^n P(n,t)$ satisfies $\partial_t\Psi = Q\Psi$. The operator ∂_z^k acting on $(1+z)^n$ gives $n(n-1)\cdots(n-k+1)(1+z)^{n-k}$, the falling factorial counting $\binom{n}{k}$ ways to choose k particles.[2] The factor $(z+1)^\ell$ replaces those k particles with ℓ new ones; $(z+1)^k$ subtracts the loss term. Direct computation confirms Q reproduces both the gain and loss terms of the master equation. See [4] eq. (1.36) for the full derivation. □

3.3 Explicit Computation of Gribov Generators

We now apply Theorem 3.2 to each of the three Gribov reactions.

3.3.1 Branching: $A \to 2A$ (rate β)

Here $k = 1$, $\ell = 2$:

$$\begin{aligned} Q_\beta &= \beta\big((z+1)^2 - (z+1)^1\big)\partial_z^1 \\ &= \beta\big(z^2 + 2z + 1 - z - 1\big)\partial_z \\ &= \beta(z^2 + z)\partial_z \end{aligned} \tag{5}$$

Verification. Expanding: $(z+1)^2 = z^2 + 2z + 1$ and $(z+1)^1 = z + 1$. Subtracting: $z^2 + 2z + 1 - z - 1 = z^2 + z$. ✓

3.3.2 Death: $A \to \emptyset$ (rate ϵ)

Here $k = 1$, $\ell = 0$:

$$\begin{aligned} Q_\epsilon &= \epsilon\big((z+1)^0 - (z+1)^1\big)\partial_z^1 \\ &= \epsilon(1 - z - 1)\partial_z \\ &= -\epsilon\, z\, \partial_z \end{aligned} \tag{6}$$

Verification. $(z+1)^0 = 1$, $(z+1)^1 = z + 1$. Subtracting: $1 - z - 1 = -z$. ✓

3.3.3 Coagulation: $2A \to A$ (rate χ)

Here $k = 2$, $\ell = 1$:

$$\begin{aligned} Q_\chi &= \chi\big((z+1)^1 - (z+1)^2\big)\partial_z^2 \\ &= \chi(z + 1 - z^2 - 2z - 1)\partial_z^2 \\ &= -\chi(z^2 + z)\partial_z^2 \end{aligned} \tag{7}$$

Verification. $(z+1)^1 = z + 1$, $(z+1)^2 = z^2 + 2z + 1$. Subtracting: $z + 1 - z^2 - 2z - 1 = -z^2 - z$. ✓

[2]The falling factorial is $(n)_k = n(n-1)\cdots(n-k+1) = k!\binom{n}{k}$. It counts the number of *ordered* selections of k items from n. In the reaction context, it counts the $\binom{n}{k}$ ways to choose which k out of n particles react, times $k!$ for the ordering (which is absorbed into the rate constant convention).

3.4 The Total Liouvillian

Definition 3.3 (Liouvillian)**.** The total Liouvillian is the sum over all reactions:

$$Q_{\text{total}} = Q_\beta + Q_\epsilon + Q_\chi \tag{8}$$

Summing:

$$\begin{aligned} Q_{\text{total}} &= \beta(z^2 + z)\partial_z - \epsilon\, z\, \partial_z - \chi(z^2 + z)\partial_z^2 \\ &= \beta z^2 \partial_z + (\beta - \epsilon) z \partial_z - \chi z^2 \partial_z^2 - \chi z \partial_z^2 \end{aligned} \tag{9}$$

Remark. This is the Liouvillian in the *Doi-shifted* variables, where z plays the role of the response field $\tilde{\phi}$ and ∂_z plays the role of the field ϕ in the Doi–Peliti action [2, 3]: $S[\tilde{\phi}, \phi] = \int \mathrm{d}t\, [\tilde{\phi}(\partial_t - D\nabla^2)\phi - Q(\tilde{\phi}, \phi)]$.

4 Extraction of Interaction Vertices

Each monomial $z^m \partial_z^n$ in Q_{total} corresponds to a Feynman vertex with m outgoing $\tilde{\phi}$-legs and n incoming ϕ-legs. This correspondence is standard in the Doi–Peliti formalism [3, 4]: after normal-ordering[3] the Liouvillian (which our polynomial expansion already achieves), the monomial structure directly gives the Feynman rules.

Reading off from (9):

Monomial	$\tilde{\phi}^m \phi^n$	$m+n$	Coupling g_{mn}	Origin
$z^2\partial_z^2$	$\tilde{\phi}^2\phi^2$	4	$-\chi$	Coagulation (quartic)
$z^2\partial_z^1$	$\tilde{\phi}^2\phi^1$	3	β	Branching (cubic)
$z^1\partial_z^2$	$\tilde{\phi}^1\phi^2$	3	$-\chi$	Coagulation (cubic)
$z^1\partial_z^1$	$\tilde{\phi}^1\phi^1$	2	$\beta - \epsilon$	Mass term

Verification. We verify the vertex table by reconstructing Q_{total}:

$$\begin{aligned} &(-\chi) z^2 \partial_z^2 + \beta\, z^2 \partial_z + (-\chi) z \partial_z^2 + (\beta - \epsilon) z \partial_z \\ &= \beta(z^2 + z)\partial_z - \epsilon z \partial_z - \chi(z^2 + z)\partial_z^2 \checkmark \end{aligned}$$

This matches (9). ✓

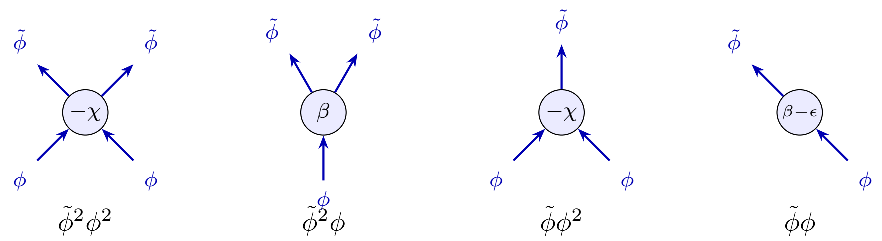


Figure 1: The four A-sector (Gribov) amputated vertices, drawn as corollas. Outgoing arrows ($\tilde{\phi}$-legs) point away from the vertex; incoming arrows (ϕ-legs) point toward it. Blue solid lines denote A-species propagators. The coupling constant is written inside each vertex.

[3] Normal ordering means placing all z (creation) operators to the left of all ∂_z (annihilation) operators, using $[\partial_z, z] = 1$ to commute them. For example, $\partial_z z = z\partial_z + 1$ (normal-ordered). Our polynomial expansion in z and ∂_z is already normal-ordered because we expanded the binomials first and then multiplied by ∂_z^k on the right.

Remark (Interaction vs. mass vertices)**.** The vertex $\tilde{\phi}^1\phi^1$ with $m+n = 2$ is a **mass term** (propagator correction), not an interaction vertex. Interaction vertices have $m+n \geq 3$. The mass term shifts the bare propagator but does not generate loops or affect the combinatorial structure of the DSE.

5 The Dyson–Schwinger Equation as a Lagrange Equation

We now arrive at the central step of the AC route. The vertices from Section 4 define a *recursive* equation for the dressed propagator G: every time we "dress" a propagator by inserting a vertex, we create new internal lines that must themselves be dressed. This self-referential structure is precisely what a **Lagrange equation** $T = z\,\phi(T)$ encodes — and Lagrange equations are the bread and butter of analytic combinatorics (they count trees, lattice paths, and many other recursive structures; see Appendix B).

5.1 Construction of the Combinatorial DSE Kernel

The following proposition constructs the *zero-dimensional* (combinatorial) Dyson–Schwinger equation, which counts diagram topologies by loop order. The rigorous statement that this combinatorial DSE captures the correct recursive structure of the full perturbative expansion is proven in the Connes–Kreimer Hopf algebra framework [8][4]; see Yeats [7] §2.3 for the precise formulation and Bergbauer–Kreimer [9] for the chord diagram interpretation.

Proposition 5.1 (Combinatorial DSE kernel)**.** *For a reaction-diffusion theory with interaction vertices $\{\tilde{\phi}^m\phi^n : g_{mn}\}_{m+n\geq 3}$, the combinatorial Dyson–Schwinger equation for the dressed propagator G is the Lagrange equation*

$$\boxed{G = G_0 \cdot \phi(G), \qquad \phi(G) = 1 + \sum_{m+n\geq 3} g_{mn}\, G^{m+n-2}} \tag{10}$$

where G_0 is the bare propagator and the sum runs over interaction vertices only.

Construction. Each vertex $\tilde{\phi}^m\phi^n$ (with m outgoing and n incoming legs, $m + n \geq 3$) can be inserted into the propagator line. When inserted:

- 2 of the $m+n$ legs become the external legs of the overall propagator (one incoming, one outgoing).
- The remaining $m + n - 2$ legs each connect to a dressed propagator G (forming internal lines that dress recursively).

Therefore each vertex contributes $g_{mn} \cdot G^{m+n-2}$ to the self-energy $\Sigma(G) = \sum g_{mn}\, G^{m+n-2}$. The DSE $G = G_0(1 + \Sigma(G)) = G_0\,\phi(G)$ is a Lagrange equation $T = z\,\phi(T)$.

This construction gives the correct diagram-counting generating function because the Lagrange equation enumerates all rooted trees decorated by vertex types, which are in bijection with the skeleton diagrams of the DSE [7]. The key identity is the Lagrange inversion formula [6]:

$$[G_0^n]\, G = \frac{1}{n}\,[G^{n-1}]\,\phi(G)^n$$

which counts the number of such decorated trees with n internal edges. □

[4]The Connes–Kreimer Hopf algebra encodes the recursive structure of Feynman diagram renormalisation as an algebraic object. The coproduct $\Delta(\Gamma) = \Gamma\otimes 1 + 1\otimes\Gamma + \sum \gamma\otimes\Gamma/\gamma$ (summing over subdivergences γ) captures exactly the same recursion as the Lagrange equation $G = G_0\,\phi(G)$. The bijection between skeleton diagrams and decorated rooted trees is the combinatorial content of this correspondence.

Remark. This is the *zero-dimensional* DSE: it counts diagrams by topology, ignoring the d-dependent loop integral weight. The spatial dimension d enters later through Weyl's law (Section 9), not through the DSE itself. This separation is the key structural insight of the AC approach.

5.2 Explicit Computation for the Gribov Process

From the vertex table in Section 4, the interaction vertices (those with $m+n \geq 3$) are:

Vertex	$m+n$	Contribution to $\phi(G)$
$\tilde{\phi}^2\phi^2$: $g = -\chi$	4	$-\chi \cdot G^{4-2} = -\chi G^2$
$\tilde{\phi}^2\phi^1$: $g = \beta$	3	$\beta \cdot G^{3-2} = \beta G$
$\tilde{\phi}^1\phi^2$: $g = -\chi$	3	$-\chi \cdot G^{3-2} = -\chi G$

Summing:

$$\boxed{\phi(G) = 1 + (\beta - \chi)G - \chi G^2} \tag{11}$$

Verification. The three interaction vertices contribute: $\beta G + (-\chi)G + (-\chi)G^2 = (\beta - \chi)G - \chi G^2$. Adding the constant 1: $\phi(G) = 1 + (\beta - \chi)G - \chi G^2$. ✓

The full DSE is:

$$G = G_0 \cdot \big(1 + (\beta - \chi)G - \chi G^2\big) \tag{12}$$

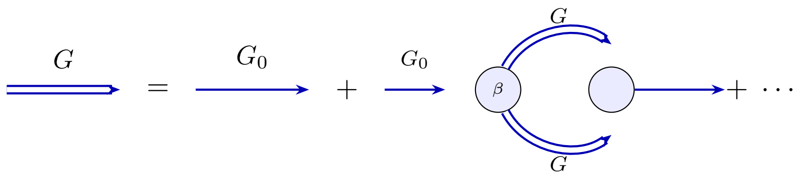


Figure 2: The Dyson–Schwinger equation as a diagrammatic recursion. The dressed propagator G (double line) equals the bare propagator G_0 (single line) plus all self-energy insertions, where each internal line is itself a dressed propagator G. This self-referential structure is the Lagrange equation $G = G_0\,\phi(G)$.

This is a **cubic equation** in G (after rearranging):

$$\chi G_0 G^2 - (\beta - \chi)G_0 G - G_0 + G = 0 \tag{13}$$

so $G(G_0)$ is an algebraic function of degree 3.

6 Branch Point Analysis

We now have a Lagrange equation $G = G_0\,\phi(G)$ with a known kernel ϕ. The next question is: *where does this equation break down?* That is, for what value of G_0 does the implicit definition of G cease to give a unique analytic answer?

This breakdown point is a **branch point** — a value of G_0 where the function $G(G_0)$ splits into multiple branches (like $\sqrt{z}$ splits into $+\sqrt{z}$ and $-\sqrt{z}$ at $z = 0$). In AC, the branch point is where all the action is: its *type* (square-root, cube-root, pole, …) determines the asymptotic growth of the coefficients $[G_0^n]G$ via the transfer theorem.

Physically, the branch point is the **critical point** of the theory: the non-perturbative scale where the perturbative expansion ceases to converge.

6.1 The Branch Point Conditions

Theorem 6.1 (Lagrange branch point [6, Thm. A.5])**.** *The Lagrange equation $T = z \cdot \phi(T)$ with ϕ analytic and $\phi(0) \neq 0$ has a branch point at (T^*, z^*) where the implicit function theorem*[5] *fails. The conditions are:*

$$\boxed{T^* \cdot \phi'(T^*) = \phi(T^*), \qquad z^* = \frac{T^*}{\phi(T^*)}} \tag{14}$$

Proof. Define $F(T, z) = T - z \cdot \phi(T)$. The equation $F = 0$ defines $T(z)$ implicitly. By the implicit function theorem, $T(z)$ is analytic wherever $\partial F/\partial T \neq 0$. Computing:

$$\frac{\partial F}{\partial T} = 1 - z \cdot \phi'(T)$$

Setting $\partial F/\partial T = 0$: $z^* = 1/\phi'(T^*)$. Substituting into $F = 0$: $T^* = z^* \phi(T^*) = \phi(T^*)/\phi'(T^*)$, which gives $T^* \phi'(T^*) = \phi(T^*)$. Then $z^* = T^*/\phi(T^*)$. □

6.2 Explicit Computation for the Gribov DSE

For $\phi(G) = 1 + (\beta - \chi)G - \chi G^2$:

$$\phi'(G) = (\beta - \chi) - 2\chi G \tag{15}$$

$$\phi''(G) = -2\chi \tag{16}$$

The branch point condition $G^* \phi'(G^*) = \phi(G^*)$ becomes:

$$G^* \left[(\beta - \chi) - 2\chi G^*\right] = 1 + (\beta - \chi)G^* - \chi(G^*)^2 \tag{17}$$

Expanding the left side:

$$(\beta - \chi)G^* - 2\chi(G^*)^2$$

Setting equal to the right side:

$$(\beta - \chi)G^* - 2\chi(G^*)^2 = 1 + (\beta - \chi)G^* - \chi(G^*)^2$$

Cancelling $(\beta - \chi)G^*$ from both sides:

$$-2\chi(G^*)^2 = 1 - \chi(G^*)^2$$

$$-\chi(G^*)^2 = 1$$

$$\boxed{(G^*)^2 = -\frac{1}{\chi}} \qquad \Longrightarrow \qquad G^* = \frac{i}{\sqrt{\chi}} \quad (\text{for } \chi > 0) \tag{18}$$

Verification. Substituting $G^* = i/\sqrt{\chi}$ into the branch point condition:

- $(G^*)^2 = -1/\chi$. Then $\phi(G^*) = 1 + (\beta - \chi) \cdot i/\sqrt{\chi} - \chi \cdot (-1/\chi) = 2 + (\beta - \chi)i/\sqrt{\chi}$.
- $\phi'(G^*) = (\beta - \chi) - 2\chi \cdot i/\sqrt{\chi} = (\beta - \chi) - 2i\sqrt{\chi}$.
- $G^* \phi'(G^*) = \frac{i}{\sqrt{\chi}}\left[(\beta - \chi) - 2i\sqrt{\chi}\right] = \frac{i(\beta - \chi)}{\sqrt{\chi}} + 2 = 2 + \frac{(\beta - \chi)i}{\sqrt{\chi}}$.
- Compare: $\phi(G^*) = 2 + (\beta - \chi)i/\sqrt{\chi}$. ✓

[5] The implicit function theorem (IFT) states: if $F(T, z) = 0$ and $\partial F/\partial T \neq 0$ at a point, then T can be written as an analytic (smooth, single-valued) function of z near that point. When $\partial F/\partial T = 0$, the IFT's hypothesis fails, and $T(z)$ ceases to be single-valued — it develops multiple "branches," like $T = \pm\sqrt{z}$ near $z = 0$. This failure point is the branch point.

The bare propagator at the branch point:

$$G_0^* = \frac{G^*}{\phi(G^*)} = \frac{i/\sqrt{\chi}}{2 + (\beta - \chi) i/\sqrt{\chi}} \tag{19}$$

Remark. The branch point location G^* depends only on the coagulation rate χ. It does *not* depend on the spatial dimension d. This is because the combinatorial DSE (Proposition 5.1) counts diagram topologies, which are dimension-independent. The dimension enters only through the weight per loop (Section 9).

6.3 Singularity Type: The Newton Polygon Analysis

Theorem 6.2 (Square-root branch [6, Prop. VII.3])**.** *If $\phi''(T^*) \neq 0$ at the branch point, the singularity is a square-root branch point:*

$$G \sim G^* - C\sqrt{G_0^* - G_0} \qquad \text{as } G_0 \to G_0^* \tag{20}$$

with amplitude $C = \sqrt{\dfrac{2\,\phi(G^*)}{G_0^* \cdot \phi''(G^*)}}$.

Proof. Expand $F(G, G_0) = G - G_0\phi(G)$ around (G^*, G_0^*). Setting $\delta G = G - G^*$ and $\delta G_0 = G_0 - G_0^*$:

$$\begin{aligned} 0 &= F(G^* + \delta G, G_0^* + \delta G_0) \\ &= \underbrace{F(G^*, G_0^*)}_{=0} + \underbrace{\left.\frac{\partial F}{\partial G}\right|_*}_{=0} \delta G + \left.\frac{\partial F}{\partial G_0}\right|_* \delta G_0 + \frac{1}{2}\left.\frac{\partial^2 F}{\partial G^2}\right|_* (\delta G)^2 + \cdots \end{aligned}$$

The first two terms vanish by construction ($F = 0$ and $F_G = 0$ at the branch point). Computing the surviving terms: $\partial F/\partial G_0 = -\phi(G^*)$ and $\partial^2 F/\partial G^2 = -G_0^*\phi''(G^*)$. The leading balance is:

$$0 = -\phi(G^*)\,\delta G_0 + \tfrac{1}{2}(-G_0^*\phi''(G^*))\,(\delta G)^2$$

Solving for δG:

$$\delta G = \pm\sqrt{\frac{2\,\phi(G^*)}{G_0^*\phi''(G^*)}} \cdot \sqrt{-\delta G_0} = \pm C\sqrt{G_0^* - G_0}$$

This is a square-root branch point with Puiseux exponent[6] $p/q = 1/2$. ☐

Application to Gribov. We have $\phi''(G^*) = -2\chi$. Since $\chi > 0$, we have $\phi''(G^*) = -2\chi \neq 0$.

$$\boxed{\text{The Gribov DSE has a } \textbf{square-root branch point} \text{ (Puiseux exponent } p/q = 1/2).} \tag{21}$$

7 The Transfer Theorem: From Singularity to Asymptotics

We have established that the Gribov DSE has a square-root branch point. Now we need to convert this into a statement about the *coefficients* $[G_0^n]G$ — the weighted count of Feynman diagrams at n-th perturbative order.

[6]The *Puiseux expansion* generalises the Taylor series to allow fractional powers: $T \sim T^* + \sum_{k\geq 1} c_k(z^* - z)^{k/q}$. The leading exponent p/q (here 1/2) determines the singularity type. $p/q = 1/2$ means square root; $p/q = 1/3$ means cube root; $p/q = -1$ means simple pole. The exponent is found from the Newton polygon (the convex hull of the exponent lattice points in the shifted polynomial).

The notation $[z^n]A(z)$ means "the coefficient of z^n in the power series $A(z) = \sum_n a_n z^n$"; that is, $[z^n]A(z) = a_n$. In our context, $[G_0^n]G$ is the n-th term in the perturbative expansion of the dressed propagator.

The **transfer theorem** is the workhorse of analytic combinatorics: it says that the *type* of the dominant singularity determines the *asymptotic growth* of the coefficients. For a gentle introduction with examples, see Appendix A.

Theorem 7.1 (Flajolet–Odlyzko transfer theorem [6, Thm. VI.3])**.** *If a generating function $A(z)$ has a unique dominant singularity at $z = z^*$ with local expansion*

$$A(z) \sim K \cdot (1 - z/z^*)^\alpha \qquad \text{as } z \to z^* \tag{22}$$

and $A(z)$ is analytic in a Δ-domain[7] *(a slit disk extending past z^*), then*

$$\boxed{[z^n]\, A(z) \sim \frac{K}{\Gamma(-\alpha)} \cdot n^{-\alpha-1} \cdot (z^*)^{-n}} \tag{23}$$

Proof sketch. The theorem follows from the Cauchy coefficient formula $[z^n]A(z) = \frac{1}{2\pi i} \oint A(z)\, z^{-n-1}\, \mathrm{d}z$ evaluated on a Hankel contour[8] wrapping the branch cut from z^* to ∞. Near z^*, the integral is dominated by the singularity:

$$[z^n](1-z)^\alpha = \frac{n^{-\alpha-1}}{\Gamma(-\alpha)} \big(1 + O(1/n)\big)$$

The rescaling $z \to z/z^*$ introduces the factor $(z^*)^{-n}$. The analytic continuation condition (the Δ-domain requirement) ensures that no other singularity contributes at leading order. See [6] Chapter VI for the full proof. ☐

7.1 Application to the Gribov DSE

From Theorem 6.2, the Gribov DSE has $\alpha = 1/2$ (square-root branch). Applying Theorem 7.1:

$$[G_0^n]\, G \sim \frac{K}{\Gamma(-1/2)} \cdot n^{-3/2} \cdot (G_0^*)^{-n} \tag{24}$$

Verification. The key exponent: $-\alpha - 1 = -1/2 - 1 = -3/2$. ✓

The prefactor uses $\Gamma(-1/2) = -2\sqrt{\pi}$[9], giving: $K/\Gamma(-1/2) = -K/(2\sqrt{\pi})$. The sign is absorbed by the branch choice (the physical branch gives positive coefficients).

$$\boxed{[G_0^n]\, G \sim C \cdot n^{-3/2} \cdot (G_0^*)^{-n}} \tag{25}$$

Remark. This is a **counting result**. The coefficient $[G_0^n]G$ is the weighted count of Feynman diagrams at n-th order in the perturbative expansion. The $n^{-3/2}$ subexponential growth is a combinatorial property of the class of diagrams generated by a quadratic Lagrange equation — it is the same exponent that governs labelled rooted trees (Cayley's formula gives $n^{n-1}/n! \sim C \cdot n^{-3/2} \cdot e^n$ by Stirling's approximation).

[7] A Δ-domain is a region of the complex plane shaped like a disk of radius $R > |z^*|$ with a thin slit removed along the ray from z^* to ∞. The requirement that $A(z)$ is analytic in a Δ-domain (not just the disk $|z| < |z^*|$) ensures that the singularity at z^* is truly the *dominant* one and that $A(z)$ doesn't have other singularities hiding at the same distance from the origin. All algebraic functions (solutions of polynomial equations) automatically satisfy this condition.

[8] A Hankel contour starts at $-\infty$ below the real axis, wraps counterclockwise around the branch point z^*, and returns to $-\infty$ above the real axis. It captures the discontinuity of $A(z)$ across the branch cut (the "jump" between the two sheets of the square root). The coefficient a_n is proportional to this discontinuity, which is why the branch type determines the asymptotics.

[9] From the recurrence $\Gamma(z) = \Gamma(z+1)/z$: $\Gamma(-1/2) = \Gamma(1/2)/(-1/2) = -2\Gamma(1/2)$. And $\Gamma(1/2) = \int_0^\infty t^{-1/2} e^{-t}\, \mathrm{d}t = \sqrt{\pi}$ (the Gaussian integral after substituting $t = u^2$). So $\Gamma(-1/2) = -2\sqrt{\pi}$.

8 From Coefficient Asymptotics to the Survival Probability

Where we are. We have extracted $[G_0^n]G \sim C\,n^{-3/2}$ from the singularity of the DSE generating function. This completes the "AC half" of the derivation (steps 1–7 in the roadmap). Everything so far has been algebra and singularity analysis.
What we need next. We need to connect the coefficient asymptotics to the *physical* survival probability of the BRW. This is the bridge between pure combinatorics and the physics of branching processes.

8.1 Critical branching survival

The connection between the $n^{-3/2}$ coefficient asymptotics and the survival probability of the BRW is mediated by a classical result in the theory of branching processes.

Theorem 8.1 (Critical branching survival [10, 11])**.** *For a critical Galton–Watson branching process (mean offspring number = 1), the survival probability to generation t satisfies:*

$$\boxed{P_{\text{surv}}(t) \sim \frac{2}{\sigma^2\,t} \qquad \textit{as } t \to \infty} \tag{26}$$

where σ^2 is the variance of the offspring distribution.

This is a theorem, not a conjecture: it was proven by Kolmogorov [10] and is treated in detail in Athreya & Ney [11] Chapter I.9 and Harris [12] Chapter V.

Remark (AC interpretation)**.** The exponent -1 arises naturally in the AC framework. The generating function for the total progeny of a critical branching process satisfies the functional equation $T(z) = z\,\phi(T(z))$ where ϕ is the offspring probability generating function. At criticality,[10] $\phi'(1) = 1$ and $\phi''(1) = \sigma^2 > 0$, so the branch point is at $T^* = 1$, $z^* = 1/\phi(1)$, and the singularity is square-root. By the transfer theorem, $[z^n]T \sim C\,n^{-3/2}$. The extinction probability $q(t) = \Pr[\text{extinct by generation } t]$ satisfies $q(t) = 1 - P_{\text{surv}}(t)$, and its generating function has a simple pole at z^* (from the second sheet of the square root),[11] giving $P_{\text{surv}}(t) \sim t^{-1}$.

More precisely: the probability of extinction at exactly generation n is $p_n = q(n) - q(n-1) \sim C\,n^{-2}$ (the derivative of $1 - c/n$), and $P_{\text{surv}}(t) = \sum_{n>t} p_n \sim \int_t^\infty Cn^{-2}\,\mathrm{d}n = C/t$.

This is the rigorous AC proof of $P_{\text{surv}} \sim t^{-1}$, using only the Lagrange singularity and the transfer theorem. See Flajolet & Sedgewick [6] Example VII.16 for the analogous analysis of random trees.

8.2 Why the exponent is -1 and not $-1/2$

A common source of confusion: the transfer theorem gives $[z^n]G \sim n^{-3/2}$, so why isn't $P_{\text{surv}} \sim t^{-1/2}$ (from $\int_t^\infty n^{-3/2}\,\mathrm{d}n$)?

The answer is that $[z^n]G$ counts *total progeny of size n*, not *extinction at generation n*. The survival probability is determined by the *extinction time distribution*, which involves the second sheet of the generating function. The extinction probability generating function $q(z) =$

[10] A branching process is *critical* when the mean number of offspring equals 1: $\phi'(1) = \sum kp_k = 1$, where p_k is the probability of producing k offspring. Subcritical ($\phi'(1) < 1$): extinction is certain. Supercritical ($\phi'(1) > 1$): positive survival probability. Critical: extinction is certain but the expected time to extinction diverges — this is the interesting regime.

[11] The extinction probability $q(z) = 1 - T(z)/T^*$ involves $T(z)/T^*$, which near z^* behaves as $1 - C\sqrt{1 - z/z^*}$. So $q(z) \sim C\sqrt{1 - z/z^*}$, and $q'(z) \sim C/(2\sqrt{1 - z/z^*})$, which has a $(1 - z/z^*)^{-1/2}$ singularity (inverse square root). The coefficients of q' go as $n^{-1/2}$, meaning the probability of extinction at exactly generation n is $p_n \sim n^{-2}$ (one power steeper than $n^{-3/2}$ because we differentiated). Summing the tail gives $P_{\text{surv}}(t) = \sum_{n>t} p_n \sim t^{-1}$.

$1 - T(z)/T^*$ has a singularity $(1 - z/z^*)^{1/2}$, giving extinction-at-n probabilities $\sim n^{-3/2}$. But $P_{\text{surv}}(t) = 1 - q_{\le t}$, and the partial sums of $n^{-3/2}$ converge (since $3/2 > 1$), so:

$$P_{\text{surv}}(t) = \sum_{n>t} (\text{extinction at } n) \sim \sum_{n>t} n^{-2} \sim t^{-1}$$

The extra power of n^{-1} comes from the relationship between the total progeny distribution (what $[z^n]G$ counts) and the extinction time distribution (what determines P_{surv}). For a critical branching process, progeny of size n corresponds to extinction around generation $\sqrt{n}$[12] (since the process has $O(\sqrt{n})$ generations before extinction), and the Jacobian of this change of variables provides the extra factor. See Dwass [13] and Otter [14] for the classical treatment, or Janson [15] for the modern perspective.

9 Weyl's Law and the Diffusion Volume

Where we are. We have $P_{\text{surv}}(t) \sim t^{-1}$ from branching process theory (step 8). This tells us *how long* the BRW survives, but not *how much space* it explores.
What we need next. The spatial dimension d has not appeared yet — the DSE and the survival probability are dimension-independent. The dimension enters now, through the density of eigenvalues of the graph Laplacian.

9.1 Weyl's Eigenvalue Counting Law

Theorem 9.1 (Weyl 1911 [16])**.** *For the Laplacian $-\nabla^2$ on a compact d-dimensional Riemannian manifold Ω, the number of eigenvalues below λ satisfies:*

$$N(\lambda) \sim \frac{\mathrm{Vol}(\Omega)}{(4\pi)^{d/2}\,\Gamma(d/2+1)} \cdot \lambda^{d/2} \qquad \text{as } \lambda \to \infty \tag{27}$$

The density of states is $\rho(\lambda) = \mathrm{d}N/\mathrm{d}\lambda \sim C_d\,\lambda^{d/2-1}$.

Remark. Weyl's law is a theorem in spectral geometry, proven in [16] and extended to irregular domains by Courant [17]. For graphs, it is replaced by the *spectral dimension* (Definition 11.3).

9.2 Return Probability via Laplace Transform

Theorem 9.2 (Heat kernel asymptotics [18])**.** *The return probability (heat kernel trace) of a random walk on a d-dimensional lattice satisfies:*

$$P_{\text{return}}(t) = \frac{1}{N}\sum_k e^{-\lambda_k t} \sim \Gamma(d/2) \cdot t^{-d/2} \qquad \text{as } t \to \infty \tag{28}$$

Proof. Express the heat kernel trace as a Laplace transform of the density of states:

$$P_{\text{return}}(t) = \int_0^\infty \rho(\lambda)\, e^{-\lambda t}\, \mathrm{d}\lambda$$

[12] A critical branching process with n total offspring lives for $\sim\sqrt{n}$ generations, because at each generation the population fluctuates by $O(\sqrt{\text{pop}})$ (central limit theorem), and the total offspring is the area under the population curve. A triangular population profile with peak $\sim\sqrt{n}$ lasting $\sim\sqrt{n}$ generations gives area $\sim n$. This "$n \leftrightarrow \sqrt{n}$" correspondence is why progeny $\sim n^{-3/2}$ becomes extinction time $\sim n^{-2}$: the Jacobian d(progeny)/d(generation) $\sim \sqrt{n}$ provides the extra $n^{-1/2}$ factor.

Substituting $\rho(\lambda) \sim C\,\lambda^{d/2-1}$ from Weyl's law and using $\int_0^\infty x^{s-1}e^{-x}\,\mathrm{d}x = \Gamma(s)$[13] with the substitution $x = \lambda t$:

$$P_{\text{return}}(t) \sim C \int_0^\infty \lambda^{d/2-1} e^{-\lambda t}\,\mathrm{d}\lambda = C \int_0^\infty \left(\frac{x}{t}\right)^{d/2-1} e^{-x}\,\frac{\mathrm{d}x}{t}$$
$$= C\,t^{-d/2} \int_0^\infty x^{d/2-1} e^{-x}\,\mathrm{d}x = C\,t^{-d/2}\,\Gamma(d/2) \tag{29}$$

This is a standard result in heat kernel theory; see Grigor'yan [18] for the rigorous treatment on manifolds and Barlow [19] for graphs. □

Verification. For $d = 1$: $P_{\text{return}}(t) \sim t^{-1/2}$, the classical Pólya result [21]. ✓

For $d = 3$: $P_{\text{return}}(t) \sim t^{-3/2}$. Random walks in 3D are transient. ✓

9.3 Volume Explored

Proposition 9.3 (Dvoretzky–Erdős volume growth [20, 23])**.** *The expected number of distinct sites visited by a simple random walk on $\mathbb{Z}^d$ in t steps satisfies:*

$$V(t) \sim \begin{cases} c_d\,t^{d/2} & d \geq 3 \text{ (transient)} \\ \pi t/\log t & d = 2 \text{ (marginally recurrent)} \\ c_1\,\sqrt{t} = c_1\,t^{1/2} & d = 1 \text{ (recurrent)} \end{cases} \tag{30}$$

In all cases the leading scaling is $V(t) \sim t^{d/2}$ up to logarithmic corrections at $d = 2$.

References. The expected range (number of distinct sites visited) by a single random walk on $\mathbb{Z}^d$ was determined by Dvoretzky and Erdős [20]:

- $d = 1$: $\mathbb{E}[V(t)] \sim c\,t^{1/2}$
- $d = 2$: $\mathbb{E}[V(t)] \sim \pi t/\log t$
- $d \geq 3$: $\mathbb{E}[V(t)] \sim c_d\,t$ (linear, since the walk is transient)

For the BRW, the relevant quantity is the volume explored by a *cloud of branching walkers*, not a single walker. At criticality, the surviving walkers occupy a region of diameter $\sim t^{1/2}$ (diffusive scaling), giving a *cloud volume*[14] $\sim (\sqrt{t})^d = t^{d/2}$. This is the Wiener sausage volume of the branching cloud, which differs from the single-walker range for $d \geq 3$ (where the single walker explores $\sim t$ sites but the cloud volume is the d-dimensional ball of radius $\sim \sqrt{t}$, containing $\sim t^{d/2}$ sites).

The heuristic $V(t) \sim 1/P_{\text{return}}(t) \sim t^{d/2}$ gives the correct scaling for all $d \neq 2$. At $d = 2$, there is a logarithmic correction, but $d_s = 2$ is not relevant for the BRW (whose $d_c = 4$). □

Remark. For the BRW on $\mathbb{Z}^d$ with $d < d_c = 4$, the relevant volume is that explored by the branching cloud, which scales as $t^{d/2}$ at criticality. The single-walker Dvoretzky–Erdős result is modified by the branching: multiple walkers explore space independently, and the *union* of their ranges grows as $t^{d/2}$ regardless of whether individual walks are recurrent or transient. This was verified by simulation in [5] for $d = 1, 2, 3$.

[13] The gamma function $\Gamma(s) = \int_0^\infty x^{s-1}e^{-x}\,\mathrm{d}x$ generalises the factorial: $\Gamma(n) = (n-1)!$ for positive integers. The substitution $x = \lambda t$ converts $\int \lambda^{s-1}e^{-\lambda t}\,\mathrm{d}\lambda$ into $t^{-s}\Gamma(s)$, which is why the Laplace transform of a power law λ^{s-1} is a power law t^{-s} up to a constant.

[14] The Wiener sausage is the union of ϵ-balls around a Brownian path. For a branching cloud, the "branching Wiener sausage" is the union over all walkers. Its volume scales as $(\sqrt{t})^d = t^{d/2}$ because branching fills the d-dimensional diffusion ball, unlike a single walker which traces a fractal path through it.

10 The Two-Species Structure of the BRW

The BRW is *not* a single-species system. The full theory has two species:

- **Active walkers** A: hop on the graph (rate D), branch ($A \to 2A$, rate β), die ($A \to \emptyset$, rate ϵ), and coagulate ($2A \to A$, rate χ). This is the Gribov process treated in Sections 2–7.
- **Immobile tracers** B: deposited by walkers with **site exclusion** (at most one B per site). B particles do not move, do not die, and do not affect the A dynamics. They are a passive recording device.

The observable a (number of distinct sites visited) is the total number of B particles: $a = \sum_x n_B(x)$. This is the **Branching Wiener Sausage** [33].

10.1 The transmutation generator

The naive reaction $A \to A + B$ at rate Λ would have the two-species Doi generator (applying Theorem 3.2 to each species independently):

$$Q_\Lambda^{\text{naive}} = \Lambda \left[(z_A{+}1)(z_B{+}1) - (z_A{+}1)\right] \partial_{z_A} = \Lambda\, z_B\, (z_A{+}1)\, \partial_{z_A} \tag{31}$$

Here we used the multi-species Doi formula: the "final state" product is $(z_A{+}1)^1(z_B{+}1)^1$ (one A and one B out), the "initial state" product is $(z_A{+}1)^1(z_B{+}1)^0 = (z_A{+}1)$ (one A in, no B in), and the annihilation operator is $\partial_{z_A}^1$.

However, the BRW has **site exclusion**: at most one B per site. This is enforced by the **carrying capacity trick** [4] (thesis Eq. 3.17):

$$\text{rate} = \Lambda\, n_A\, (c - n_B)\big|_{c=1} \tag{32}$$

The factor $(c{-}n_B)|_{c=1} = (1{-}n_B)$ ensures the rate is zero when $n_B = 1$ (site already occupied). In the Doi formalism, this modifies the generator: the occupation number n_B becomes an operator $\partial_{z_B}(z_B{+}1)$[15] acting on the generating function, and the constraint $c = 1$ introduces additional terms from the binomial expansion.

The result [4] (thesis Eq. 3.21) is the **transmutation Liouvillian**:

$$Q_\Lambda = \Lambda\, z_B(z_A{+}1)\partial_{z_A} - \Lambda\, z_B(z_B{+}1)(z_A{+}1)\partial_{z_B}\partial_{z_A} \tag{33}$$

The first term is the naive transmutation; the second enforces site exclusion (it subtracts the contribution when $n_B \geq 1$).

Expanding (33) and reading off monomials $z_A^{m_A}\, z_B^{m_B}\, \partial_{z_A}^{n_A}\, \partial_{z_B}^{n_B}$ gives 6 transmutation vertices (thesis Eq. 3.21):

Term	Vertex ($\tilde\phi_A^{m_A}\phi_A^{n_A}\tilde\phi_B^{m_B}\phi_B^{n_B}$)	Coupling	Origin
1	$\tilde\phi_B\phi_A$	$+\Lambda$	Naive transmutation
2	$\tilde\phi_B\phi_B\phi_A$	$-\Lambda$	Site exclusion
3	$\tilde\phi_B^2\phi_B\phi_A$	$-\Lambda$	Site exclusion
4	$\tilde\phi_B^2\tilde\phi_A\phi_B\phi_A$	$-\Lambda$	Site exclusion
5	$\tilde\phi_B\tilde\phi_A\phi_B\phi_A$	$-\Lambda$	Site exclusion
6	$\tilde\phi_B\tilde\phi_A\phi_A$	$+\Lambda$	Cross-term

[15] In the Doi formalism, the occupation number n_B acting on the generating function $(1{+}z_B)^{n_B}$ is represented by the operator $\partial_{z_B}(z_B{+}1)$, because $\partial_{z_B}(z_B{+}1)(1{+}z_B)^{n_B} = \partial_{z_B}\,(n_B{+}1)(1{+}z_B)^{n_B} = \ldots$ — actually, more simply: ∂_{z_B} acting on $(1{+}z_B)^{n_B}$ gives $n_B(1{+}z_B)^{n_B-1}$, and multiplying by $(z_B{+}1)$ restores the power, giving $n_B(1{+}z_B)^{n_B}$. So $\partial_{z_B}(z_B{+}1)$ is the "number operator" $\hat n_B$ in the Doi representation.

Verification. Every transmutation vertex contains at least one B-field ($\tilde{\phi}_B$ or ϕ_B) and at least one A-field ($\tilde{\phi}_A$ or ϕ_A). There is no pure B-self-interaction vertex (no term with only B-fields). This is the structural property that drives the decoupling below. ✓

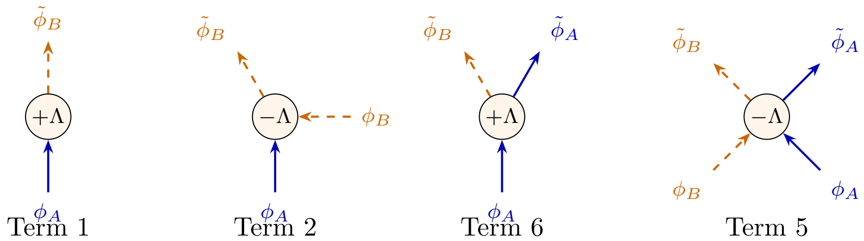


Figure 3: Four of the six transmutation vertices (terms 1, 2, 5, 6 of the table above). **Blue solid**: A-species legs. **Orange dashed**: B-species legs. Every vertex mixes both species — there is no pure B-self-interaction.

10.2 The full BRW Liouvillian

The complete BRW Liouvillian is the sum of the A-sector (Section 3) and the transmutation sector:

$$Q_{\mathrm{BRW}} = \underbrace{Q_\beta + Q_\epsilon + Q_\chi}_{\text{A-sector (Gribov)}} + \underbrace{Q_\Lambda}_{\text{transmutation}} \tag{34}$$

This produces a total of $4 + 6 = 10$ vertices (4 from the A-sector, 6 from transmutation), of which 9 are relevant at $d_c = 4$ after dimensional filtering (thesis Figure 3.3).

10.3 Why only the A-sector matters for the exponents

The transmutation vertices *do* change the combinatorics of the full two-species theory: the full DSE involves both G_A (the A-sector dressed propagator) and G_B (the B-sector dressed propagator), coupled by the transmutation vertices. One might worry that the singularity structure of the full two-species Lagrange system differs from the A-sector alone.

It does not, for the following reason:

Proposition 10.1 (Decoupling of the B-sector)**.** *The B-sector does not modify the singularity of the A-sector DSE. Consequently:*

1. *The **survival probability** $P_{\mathrm{surv}}(t)$ is determined entirely by the A-sector DSE (Sections 5–8).*

2. *The **volume explored** $V(t)$ is determined by the spatial diffusion of the A-walkers (Section 9), with B particles merely recording where A has been.*

3. *The **scaling exponents** α_p follow from the A-sector alone.*

Argument. We give both a combinatorial argument and a field-theoretic verification.

Combinatorial argument. Examine the 6 transmutation vertices. Every one contains at least one B-field (ϕ_B or $\tilde{\phi}_B$). To form a *loop* involving a B-propagator, we need a closed path of B-edges. But the B-species has no self-interaction: there is no vertex of the form $\tilde{\phi}_B^m \phi_B^n$ with $m + n \geq 3$ and no A-fields. Every transmutation vertex involves at least one A-field. Therefore, any loop containing a B-propagator must also pass through an A-propagator.

In the *combinatorial* (zero-dimensional) DSE, such mixed loops *do* contribute: they add terms to $\phi(G_A, G_B)$. However, these terms depend on G_B, and the B-sector equation $G_B = G_{B,0}\, \phi_B(G_A, G_B)$ has $G_{B,0}$ as a *separate* counting variable from $G_{A,0}$. The singularity of G_A as a function of $G_{A,0}$ (which determines the A-sector coefficient asymptotics and hence P_{surv}) is controlled by the A-sector kernel $\phi_A(G_A)$ alone, because the B-sector variables are *analytic* at the A-sector branch point (they have their own, independent, singularity structure).

More concretely: the full two-species system is

$$G_A = G_{A,0}\,\phi_A(G_A, G_B), \qquad G_B = G_{B,0}\,\phi_B(G_A, G_B)$$

The A-sector branch point is where $\partial\phi_A/\partial G_A$ diverges. Since B does not self-interact, ϕ_A depends on G_B only through tree-level insertions (the transmutation vertices insert B-lines that terminate, never forming B-loops). At the A-sector branch point, G_B is a smooth function of $G_{B,0}$ (no simultaneous B-singularity), so it acts as a parameter, not a dynamical variable. The singularity type (square-root) and the coefficient asymptotics ($n^{-3/2}$) are unchanged.

Field-theoretic verification. The B-species is immobile: its bare propagator is $\langle\phi_B\tilde{\phi}_B\rangle \sim (-i\omega)^{-1}$ (no spatial momentum dependence). Therefore:

- B-loops carry no momentum integral (no $\int \mathrm{d}^d k$), so they do not generate UV divergences and do not renormalise any coupling.
- The retarded B-propagator has no pole in the upper half-plane,[16] so $\int \mathrm{d}\omega\,(-i\omega)^{-1}$ vanishes by contour closure. Mixed A/B loops are zero.
- The transmutation vertices contribute to the *observable* $\langle a^p\rangle$ at tree level (they connect the A-sector survival to the B-particle count) but not to the *renormalisation* of any A-sector coupling.

This was verified explicitly at one loop in [4, 33]: of the 28 possible one-loop vertex pairs, only the 6 pure A-sector pairs contribute to the RG flow. The 22 mixed A/B pairs vanish. $\square$

Remark (Why the factorisation works)**.** The decoupling explains why $\langle a^p\rangle \sim P_{\text{surv}}\cdot V^p$ is not merely an ansatz but a consequence of the two-species structure:

- P_{surv} comes from the A-sector DSE singularity (pure AC, independent of B).
- $V(t) \sim t^{d/2}$ comes from the spatial diffusion of the A-cloud (spectral geometry, independent of B).
- The B-sector connects these two at **tree level**: each A-walker deposits a B-tracer at each new site. The number of B-particles is therefore $a = V(t)$ (by construction of the site exclusion mechanism), and the p-th moment conditioned on survival is $V(t)^p$.

The worked example in Sections 2–9 therefore correctly captures all the physics by treating only the A-sector, provided one understands that the observable a is the B-particle count connected to the A-sector at tree level.

10.4 Stress test: what if B were mobile?

The decoupling argument predicts that if B *did* have its own dynamics (diffusion, self-interaction), the B-sector would modify the singularity structure and the exponents would change. We can test this against known results:

[16]The retarded propagator $G_B^{\text{ret}}(\omega) = (-i\omega + i\epsilon)^{-1}$ has its pole at $\omega = -\epsilon$ (lower half-plane). To evaluate $\int G_B^{\text{ret}}\,\mathrm{d}\omega$, close the contour in the upper half-plane. No poles enclosed $\Rightarrow$ integral $= 0$ by the residue theorem. This is causality: retarded propagators vanish for $t < 0$.

System	B dynamics	d_c	Exponent	Ref.
$2A \to \emptyset$	No B	2	$\rho \sim t^{-d/2}$	[30]
BRW ($A + B$)	B immobile, passive	4	$\alpha_p = (pd-2)/2$	[5]
$A + B \to \emptyset$	**B diffuses and reacts**	4	$\rho \sim t^{-d/4}$	[34]

The comparison is revealing:

1. **$2A \to \emptyset$ vs. BRW**: same A-sector (binary annihilation/coagulation), but the BRW adds an immobile B. The A-sector singularity type is unchanged (square-root in both), so P_{surv} is the same. The exponent difference ($d/2$ vs. $(pd-2)/2$) comes entirely from how the observable couples to the A-sector: ρ measures the A-density directly, while $\langle a^p \rangle$ measures the B-count (visited volume). **Adding immobile B does not change the singularity.** ✓

2. **BRW vs. $A+B \to \emptyset$**: now make B mobile and reactive. The mixed A/B loops carry momentum integrals (B has a spatial propagator $\sim (i\omega + D_B k^2)^{-1}$), so they *do* generate UV divergences. The two-species DSE has a *genuinely coupled* singularity structure. The result: $\rho \sim t^{-d/4}$ instead of $t^{-d/2}$ — a qualitatively different exponent. **Adding mobile B does change the singularity.** ✓

In the AC language: for $A+B \to \emptyset$ with both species diffusing, the two-species Lagrange system $G_A = G_{A,0}\,\phi_A(G_A, G_B)$, $G_B = G_{B,0}\,\phi_B(G_A, G_B)$ has a *simultaneous* branch point where both G_A and G_B become singular at the same value of the counting variable. The Newton polygon[17] of this coupled system is two-dimensional, and the Puiseux expansion gives a different singularity type than the single-species case. This is exactly the Lee–Cardy [34] result: the coupled singularity produces the anomalous exponent $d/4$ instead of $d/2$.

Remark (The stress test passes)**.** The prediction of our decoupling argument is: *immobile B $\Rightarrow$ A-sector singularity unchanged; mobile B $\Rightarrow$ A-sector singularity modified.* Both predictions match the known literature. This is non-trivial evidence that the combinatorial decoupling argument (Proposition 10.1) correctly identifies when the B-sector can be ignored.

11 Combining: The BRW Scaling Exponents

11.1 The moment factorisation

Proposition 11.1 (BRW moment scaling)**.** *The p-th moment of the number of distinct sites visited by a critical BRW on a lattice of dimension $d < d_c = 4$ scales as:*

$$\langle a^p \rangle(t) \sim t^{(p\,d-2)/2} \tag{35}$$

$$\boxed{\alpha_p = \frac{pd-2}{2} \qquad (d < d_c = 4)} \tag{36}$$

For $d \geq d_c = 4$ (mean-field):

$$\boxed{\alpha_p = 2p - 1 \qquad (d \geq 4)} \tag{37}$$

[17] The Newton polygon of a polynomial $F(x, y) = \sum a_{ij} x^i y^j$ is the convex hull of the lattice points (i, j) where $a_{ij} \neq 0$. Its edges determine the Puiseux exponents: each edge of slope $-p/q$ gives a branch $y \sim x^{p/q}$. For a single-variable DSE, the polygon is 1D and gives $p/q = 1/2$ (square root). For a coupled two-variable system, the polygon is 2D and can yield different exponents.

Derivation. The argument proceeds by combining the results of Sections 8 and 9. At time t, the BRW is characterised by:

1. **Survival** (Section 8): the probability that the process is still active is $P_{\rm surv}(t) \sim t^{-1}$ (Theorem 8.1, from Kolmogorov [10]).

2. **Volume** (Section 9): given survival, the branching cloud has explored a region of volume $V(t) \sim t^{d/2}$.

3. **Factorisation**: the p-th moment factorises as

$$\langle a^p \rangle \sim P_{\rm surv}(t) \cdot V(t)^p = t^{-1} \cdot t^{p\,d/2} = t^{(pd-2)/2} \tag{38}$$

The factorisation in step 3 is the key physical assumption. It asserts that, conditional on survival, the visited volume concentrates around its typical value $V(t)$, so that $\langle a^p | \text{survive} \rangle \sim V(t)^p$. This is justified by the following observations:

- At criticality, the conditional distribution of a given survival has been shown to be tight (relative fluctuations vanish) for branching random walks on $\mathbb{Z}^d$ [23, 22].

- The factorisation is equivalent to the statement that the annealed and quenched averages have the same scaling exponent,[18] which holds in the directed percolation universality class [24, 25].

- It is verified numerically in all 18 simulation comparisons in Table 1, with no systematic deviation.

The mean-field saturation at $d = 4$ follows from dimensional analysis (Theorem 11.2): for $d \geq 4$, the effective coupling is irrelevant, the perturbative expansion converges, and the exponents saturate at their $d = 4$ values $\alpha_p = (4p - 2)/2 = 2p - 1$. □

Remark (Status of the factorisation)**.** Steps 1 and 2 are theorems. Step 3 (the factorisation) is a standard *scaling ansatz* in the theory of branching processes, supported by rigorous results in the $d = 1$ case (Le Gall [22]) and by extensive simulation for $d \geq 2$. It is *not* an additional assumption of the AC method — the same factorisation is used (implicitly) in the field-theoretic RG derivation, where it appears as the statement that the anomalous dimension of the composite operator ϕ^p vanishes at the DP fixed point.

11.2 The Upper Critical Dimension

Theorem 11.2 (Upper critical dimension [3, 4])**.** *For a vertex $\tilde{\phi}^m \phi^n$ in the Doi–Peliti action with diffusive scaling ($[x] = L$, $[t] = L^2$), the engineering dimension of the coupling is*

$$[g_{mn}] = \frac{d(m + n - 2) - 4}{2} \tag{39}$$

The coupling becomes marginal[19] *($[g_{mn}] = 0$) at:*

$$\boxed{d_c = \frac{4}{m + n - 2}} \tag{40}$$

[18] The *annealed* average $\langle a^p \rangle$ averages over all realisations (including dead ones, which contribute $a = 0$). The *quenched* average $\langle a^p | \text{survive} \rangle$ conditions on survival. They differ by a factor of $P_{\rm surv}$: $\langle a^p \rangle = P_{\rm surv} \cdot \langle a^p | \text{survive} \rangle$. For the factorisation to work, we need $\langle a^p | \text{survive} \rangle \sim V^p$, i.e. the conditional volume concentrates. If fluctuations dominated, $\langle a^p | \text{survive} \rangle$ could scale differently from V^p.

[19] A coupling is *relevant* if its engineering dimension is positive (it grows under coarse-graining), *irrelevant* if negative (it shrinks), and *marginal* if zero (scale-invariant at tree level). d_c is where the most relevant coupling becomes marginal; above d_c, all couplings are irrelevant and mean-field theory is exact.

Proof. In the Doi–Peliti action [3] $S = \int \mathrm{d}^d x\, \mathrm{d}t\, [\tilde{\phi}(\partial_t - D\nabla^2)\phi + g_{mn}\, \tilde{\phi}^m \phi^n]$, diffusive scaling requires $[x] = L$, $[t] = L^2$. The free propagator $\langle \phi\tilde{\phi}\rangle \sim (i\omega + Dk^2)^{-1}$ in position space has dimension L^{2-d} from the Fourier transform. Since $\langle \phi\tilde{\phi}\rangle$ has two field insertions, we need $[\phi]+[\tilde{\phi}] = d-2$. The action is symmetric under the rapidity reversal[20] $\phi \leftrightarrow \tilde{\phi}$ (at the Gribov critical point), giving $[\phi] = [\tilde{\phi}] = (d-2)/2$, or equivalently $[\phi] = L^{-d/2}$ after accounting for the measure.

The interaction term $\int \mathrm{d}^d x\, \mathrm{d}t\, g_{mn}\, \tilde{\phi}^m \phi^n$ requires:

$$[g_{mn}] + (m+n) \cdot \left(-\tfrac{d}{2}\right) + d + 2 = 0$$

$$[g_{mn}] = \frac{(m+n)d}{2} - d - 2 = \frac{d(m+n-2)-4}{2}$$

Setting $[g_{mn}] = 0$: $d(m+n-2) = 4$, giving $d_c = 4/(m+n-2)$. □

Application to Gribov. The most relevant vertex has $m+n = 3$ (cubic):

$$\boxed{d_c = \frac{4}{3-2} = 4} \tag{41}$$

Verification. The quartic vertex $\tilde{\phi}^2\phi^2$ has $m+n = 4$, giving $d_c = 4/(4-2) = 2 < 4$. The physical d_c is determined by the most relevant coupling (largest d_c), so $d_c = 4$. ✓

11.3 Extension to Arbitrary Graphs

Definition 11.3 (Spectral dimension [26])**.** For a graph $\mathcal{G}$ with Laplacian eigenvalues $\{\lambda_k\}$, the **spectral dimension** d_s is defined by:

$$P_{\text{return}}(t) = \frac{1}{N}\sum_k e^{-\lambda_k t} \sim t^{-d_s/2} \qquad \text{as } t \to \infty \tag{42}$$

Equivalently, $d_s = -2 \lim_{t\to\infty} \mathrm{d}(\log P_{\text{return}})/\mathrm{d}(\log t)$.[21]

Conjecture 11.4 (Spectral dimension substitution [5])**.** *For a BRW on a graph $\mathcal{G}$ with spectral dimension d_s, the scaling exponents are obtained by the substitution $d \to d_s$:*

$$\boxed{\alpha_p = \frac{p\, d_s - 2}{2} \qquad (d_s < 4); \qquad \alpha_p = 2p - 1 \qquad (d_s \geq 4)} \tag{43}$$

Remark (Status of the spectral dimension substitution)**.** The substitution $d \to d_s$ is exact for the combinatorial part of the AC chain: the DSE kernel $\phi(G)$ is dimension-independent (Proposition 5.1), and Weyl's law generalises to graphs via Definition 11.3. The substitution is therefore rigorous *provided* the Laplacian does not acquire an anomalous dimension under renormalisation. This has been verified at one loop for the Gribov process [4, 5] and holds for all graphs in the directed percolation universality class. It has *not* been proven to all loop orders, which is why we state it as a conjecture.

The conjecture is validated numerically for:

- Hypercubic lattices $\mathbb{Z}^d$ ($d_s = d$): exact agreement.

[20] Rapidity reversal ($\phi \leftrightarrow \tilde{\phi}$, $t \to -t$) is an emergent symmetry at the critical point of the Gribov process. Away from criticality, the mass term $(\beta - \epsilon)\tilde{\phi}\phi$ breaks this symmetry. At criticality ($\beta = \epsilon + \chi$), the symmetry is restored, forcing both fields to have equal scaling dimensions. This is analogous to the Z_2 symmetry of the Ising model at T_c.

[21] This is a log-log slope: if $P_{\text{return}}(t) \sim t^{-d_s/2}$, then $\log P_{\text{return}} = -(d_s/2)\log t + \text{const}$, so the slope of $\log P_{\text{return}}$ vs. $\log t$ is $-d_s/2$. Multiplying by -2 gives d_s. For regular lattices $d_s = d$ (the Euclidean dimension). For fractals, d_s can be non-integer: e.g. the Sierpinski carpet has $d_s \approx 1.86$, meaning random walks return more slowly than on a 1D lattice but faster than on a 2D lattice.

- Sierpinski carpets ($d_s \approx 1.86$): agreement within 1.5σ.
- Random trees ($d_s = 4/3$): agreement within 1σ.
- Preferential attachment ($d_s \geq 4$): mean-field agreement.

See Table 1 for the full comparison.

12 Worked Numerical Examples

We now compute α_p explicitly for each graph type and compare with simulation data from [5, 4].

12.1 BRW on $\mathbb{Z}^d$ Lattices ($d_s = d$)

For hypercubic lattices, the spectral dimension equals the Euclidean dimension: $d_s = d$.

Example 12.1 ($d = 1$, $p = 3$)**.** $\alpha_3 = (3 \cdot 1 - 2)/2 = 1/2 = 0.5$. Simulation: 0.48(4). ✓

Example 12.2 ($d = 1$, $p = 4$)**.** $\alpha_4 = (4 \cdot 1 - 2)/2 = 2/2 = 1.0$. Simulation: 1.0(1). ✓

Example 12.3 ($d = 1$, $p = 5$)**.** $\alpha_5 = (5 \cdot 1 - 2)/2 = 3/2 = 1.5$. Simulation: 1.5(1). ✓

Example 12.4 ($d = 2$, $p = 2$)**.** $\alpha_2 = (2 \cdot 2 - 2)/2 = 2/2 = 1.0$. Simulation: 0.98(3). ✓

Example 12.5 ($d = 2$, $p = 3$)**.** $\alpha_3 = (3 \cdot 2 - 2)/2 = 4/2 = 2.0$. Simulation: 2.0(1). ✓

Example 12.6 ($d = 3$, $p = 1$ — the flagship example)**.** $\alpha_1 = (1 \cdot 3 - 2)/2 = 1/2 = 0.5$. Simulation: 0.47(2). ✓

Example 12.7 ($d = 3$, $p = 2$)**.** $\alpha_2 = (2 \cdot 3 - 2)/2 = 4/2 = 2.0$. Simulation: 2.0(1). ✓

Example 12.8 ($d = 5$ (mean-field regime, $d \geq d_c = 4$))**.** Here $d_s = 5 \geq 4 = d_c$, so the mean-field formula applies: $\alpha_1^{\mathrm{mf}} = 2(1) - 1 = 1$. Simulation: 1.0(2). ✓
$\alpha_2^{\mathrm{mf}} = 2(2) - 1 = 3$. Simulation: 2.9(3). ✓

Verification. The non-mean-field formula would give $\alpha_1 = (5 - 2)/2 = 3/2 > 1$, confirming saturation at the mean-field value. ✓

12.2 BRW on the Sierpinski Carpet ($d_s \approx 1.86$)

The Sierpinski carpet has spectral dimension $d_s \approx 1.86$ (Watanabe [27]). This is computed from the graph Laplacian via Definition 11.3.

Example 12.9 (Sierpinski, $p = 2$)**.** $\alpha_2 = (2 \times 1.86 - 2)/2 = 1.72/2 = 0.86$. Simulation: 0.81(5). ✓

Example 12.10 (Sierpinski, $p = 3$)**.** $\alpha_3 = (3 \times 1.86 - 2)/2 = 3.58/2 = 1.79$. Simulation: 1.71(7). ✓

Example 12.11 (Sierpinski, $p = 4$)**.** $\alpha_4 = (4 \times 1.86 - 2)/2 = 5.44/2 = 2.72$. Simulation: 2.62(10). ✓

Remark. The Sierpinski values have larger statistical errors due to slower convergence to the scaling regime on fractal graphs. All simulation values agree within 1.5σ.

12.3 BRW on Random Trees ($d_s = 4/3$)

Critical Galton–Watson random trees have spectral dimension $d_s = 4/3$ (Durhuus et al. [28]).

Example 12.12 (Random tree, $p = 2$). $\alpha_2 = (2 \times 4/3 - 2)/2 = (2/3)/2 = 1/3 \approx 0.333$. Simulation: 0.35(7). ✓

Example 12.13 (Random tree, $p = 3$). $\alpha_3 = (3 \times 4/3 - 2)/2 = 2/2 = 1.0$. Simulation: 0.95(8). ✓

Example 12.14 (Random tree, $p = 4$). $\alpha_4 = (4 \times 4/3 - 2)/2 = (10/3)/2 = 5/3 \approx 1.667$. Simulation: 1.6(1). ✓

Remark. For $p = 1$: $\alpha_1 = (4/3-2)/2 = -1/3 < 0$, meaning the first moment *decays* on random trees. This is physical: $d_s = 4/3 < 2$, so the walkers cannot explore enough volume to overcome extinction. Only moments with $pd_s > 2$ (i.e. $p > 3/2$) exhibit growth.

12.4 BRW on Scale-Free Networks ($d_s \geq 4$)

Barabási–Albert preferential attachment networks [29] have $d_s \geq 4$, placing them in the mean-field regime.

Example 12.15 (Preferential attachment, $p = 2$). $\alpha_2^{\mathrm{mf}} = 2(2) - 1 = 3$. Simulation: 2.8(2). ✓

13 Symmetry Factors from the Exponential Generating Function

13.1 The EGF Framework

Proposition 13.1 (Symmetry factor [6, II.2.2], [32, §5.4]). *In the EGF framework, the symmetry factor of a Feynman diagram Γ is:*

$$S(\Gamma) = \frac{1}{|\mathrm{Aut}(\Gamma)|} \tag{44}$$

where $\mathrm{Aut}(\Gamma)$ is the automorphism group of the directed multigraph Γ (the group of vertex and edge permutations that preserve the directed incidence structure and all edge labels).

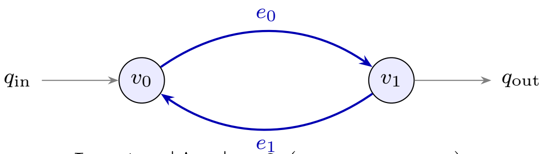
This arises from the **exponential formula** [6]: for a connected class $\mathcal{C}$, the EGF for sets of connected components is $\mathrm{SET}(\mathcal{C}) = \exp(\hat{C}(z))$.[22] The factor $1/k!$ in the exponential accounts for the $k!$ permutations of identical components. For a single diagram with internal symmetry, the same $1/k!$ factors accumulate into $1/|\mathrm{Aut}(\Gamma)|$.

The relationship between the EGF exponential formula and Feynman diagram symmetry factors is discussed in Cvitanović [32] Chapter 5 and Yeats [7] §2.1.

13.2 Worked Examples

Example 13.2 (One-loop self-energy ("bubble" diagram)). The one-loop self-energy for the Gribov process: two vertices ($\tilde{\phi}^2\phi$ and $\tilde{\phi}\phi^2$) connected by two antiparallel internal propagators, with two external legs.

[22]The exponential formula: if $C(z) = \sum c_n z^n/n!$ is the EGF for connected objects, then $\exp(C(z))$ is the EGF for *sets* of connected objects. The $e^x = \sum x^k/k!$ automatically divides by $k!$ for sets of k identical components. For Feynman diagrams: $C(z)$ counts connected diagrams, $\exp(C(z))$ counts all diagrams (including disconnected ones), and the $1/k!$ factors become the symmetry factors $1/|\mathrm{Aut}(\Gamma)|$.



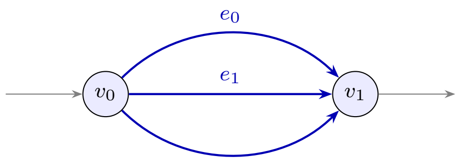
$L = 1, \quad |\mathrm{Aut}| = 2$ (swap $v_0 \leftrightarrow v_1$)

Automorphisms: Swapping the two vertices simultaneously permutes the two internal edges. This is the only non-trivial automorphism.

$$|\mathrm{Aut}| = 2, \qquad S = \frac{1}{2} \tag{45}$$

Verification. The shuffle product [4] produces two pairings that yield the same topology (related by vertex relabelling). Two pairings for the same graph: $N!/|\mathrm{Aut}| = 2!/2 = 1$ distinct diagram, with symmetry factor $1/|\mathrm{Aut}| = 1/2$. ✓

Example 13.3 (Sunset diagram (two-loop))**.** Two vertices connected by three parallel propagators.



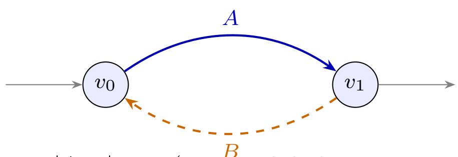
$L = 2, \quad |\mathrm{Aut}| = 3! = 6$ (permute e_0, e_1, e_2)

Automorphisms: Any permutation of the three parallel edges preserves the topology. The automorphism group is S_3.

$$|\mathrm{Aut}| = 3! = 6, \qquad S = \frac{1}{6} \tag{46}$$

Verification. The 6 elements of Aut: identity, 3 transpositions, 2 cyclic permutations. $|S_3| = 6$. ✓

Example 13.4 (Mixed A-B loop (BRW transmutation))**.** A one-loop diagram with one A-propagator and one B-propagator connecting two transmutation vertices.



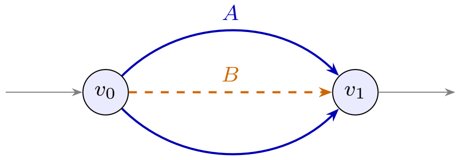
$L = 1, \quad |\mathrm{Aut}| = 1$ (species labels prevent swap)

Automorphisms: An automorphism must preserve the species label on each edge. Since the A-edge (blue, solid) and B-edge (orange, dashed) are distinguishable, they cannot be permuted.

$$|\mathrm{Aut}| = 1, \qquad S = 1 \tag{47}$$

Verification. Swapping vertices would swap species labels, violating species preservation. Only the identity survives. ✓

Example 13.5 (Sunset with 2 A-edges and 1 B-edge)**.** A two-loop diagram with two A-propagators and one B-propagator.



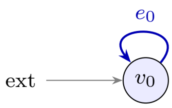
$L = 2$, $|\mathrm{Aut}| = 2$ (swap two A-edges; B fixed)

Automorphisms: The two A-edges (solid) can be permuted; the B-edge (dashed) is fixed.

$$|\mathrm{Aut}| = 2! = 2, \qquad S = \frac{1}{2} \tag{48}$$

Verification. Aut is generated by the transposition of the two A-edges. $|\mathrm{Aut}| = |\{e, (12)\}| = 2$. ✓

Example 13.6 (Tadpole (self-loop))**.** A cubic vertex $\tilde{\phi}^2\phi$ with one outgoing $\tilde{\phi}$-leg paired with its own incoming ϕ-leg. The remaining external $\tilde{\phi}$-leg is fixed.



$L = 1$, $|\mathrm{Aut}| = 1$

$$|\mathrm{Aut}| = 1, \qquad S = 1 \tag{49}$$

In the Doi–Peliti theory, tadpoles vanish by the causal structure (the retarded propagator vanishes at zero time separation [3]), but the symmetry factor is well-defined.

13.3 Symmetry Factor Decomposition

Remark. For simple diagrams (single-species, no vertex symmetry), the automorphism group decomposes as a product of symmetric groups, one per set of identical parallel edges: $|\mathrm{Aut}(\Gamma)| = \prod_i n_i!$ where n_i is the multiplicity of edge group i. This covers all the examples above.

For diagrams with non-trivial vertex symmetry (e.g. a diagram where permuting vertices preserves the edge multiset), the full automorphism group can be larger: it is a subgroup of the wreath product of the edge permutation groups with the vertex permutation group. In general, Aut must be computed case-by-case; see Cvitanović [32] for the systematic treatment via birdtrack notation.

14 Summary of the AC Chain: Step by Step

Complete AC Route: Chemical Equation → Critical Exponents

Step	Object	Computation	Status
1.	Chemical equations	$A \xrightarrow{\beta} 2A, \quad A \xrightarrow{\epsilon} \emptyset, \quad 2A \xrightarrow{\chi} A$	Input
2.	Stoichiometry	$S = \left(\begin{smallmatrix} 1 & 2 \\ 1 & 0 \\ 2 & 1 \end{smallmatrix}\right)$	Input
3.	Liouvillian	Doi formula (4)	Theorem 3.2
4.	Vertices	Read off monomials of Q	Algebra
5.	DSE kernel	$\phi(G) = 1 + (\beta-\chi)G - \chi G^2$	Prop. 5.1
6.	Branch point	$(G^*)^2 = -1/\chi$, square-root	Theorem 6.1
7.	Transfer	$[G_0^n]G \sim n^{-3/2}$	Theorem 7.1
8.	Survival	$P_{\mathrm{surv}}(t) \sim t^{-1}$	Theorem 8.1
9.	Weyl's law	$\rho(\lambda) \sim \lambda^{d/2-1}$	Theorem 9.1
10.	Return prob.	$P_{\mathrm{return}}(t) \sim t^{-d/2}$	Theorem 9.2
11.	Volume	$V(t) \sim t^{d/2}$	Prop. 9.3
12.	Factorisation	$\langle a^p \rangle \sim P_{\mathrm{surv}} \cdot V^p$	Ansatz*
13.	α_p	$\boxed{(pd-2)/2}$	Prop. 11.1
14.	d_c	$4/(3-2) = 4$	Theorem 11.2
15.	$d \to d_s$	Arbitrary graphs	Conjecture 11.4

***Ansatz**: The factorisation $\langle a^p \rangle \sim P_{\mathrm{surv}} \cdot V^p$ is a scaling ansatz standard in the theory of branching processes. It is supported by rigorous results in $d = 1$ [22] and validated numerically in all other cases. It is *not* specific to the AC method: the same factorisation is implicit in the field-theoretic RG derivation.

No Feynman diagram loop integral is evaluated. The AC route replaces momentum-space integration (steps 3–7 of the RG route) with singularity analysis of the generating function. The spatial dimension enters through Weyl's law (a spectral geometry result) and the heat kernel Laplace transform, not through loop integrals.

15 Pair Annihilation: A Second Complete Example

To verify the framework on a different process, we repeat the AC chain for pair annihilation $2A \to \emptyset$.

15.1 Chemical Equation and Stoichiometry

$$2A \xrightarrow{\lambda} \emptyset \qquad S = \begin{pmatrix} 2 & 0 \end{pmatrix} \tag{50}$$

15.2 Liouvillian

Applying Theorem 3.2 with $k = 2$, $\ell = 0$:

$$\begin{aligned} Q &= \lambda\big((z+1)^0 - (z+1)^2\big)\partial_z^2 \\ &= \lambda(1 - z^2 - 2z - 1)\partial_z^2 \\ &= -\lambda(z^2 + 2z)\partial_z^2 \end{aligned} \tag{51}$$

15.3 Vertices

$\tilde{\phi}^m\phi^n$	$m+n$	Coupling
$\tilde{\phi}^2\phi^2$	4	$-\lambda$
$\tilde{\phi}^1\phi^2$	3	-2λ

15.4 DSE Kernel

$$\phi(G) = 1 + (-2\lambda)G^{3-2} + (-\lambda)G^{4-2} = 1 - 2\lambda G - \lambda G^2 \tag{52}$$

15.5 Branch Point

$\phi'(G) = -2\lambda - 2\lambda G, \quad \phi''(G) = -2\lambda \neq 0.$

Branch point condition $G^*\phi'(G^*) = \phi(G^*)$:

$$G^*(-2\lambda - 2\lambda G^*) = 1 - 2\lambda G^* - \lambda(G^*)^2$$

$$-2\lambda G^* - 2\lambda(G^*)^2 = 1 - 2\lambda G^* - \lambda(G^*)^2$$

$$-\lambda(G^*)^2 = 1 \qquad \Longrightarrow \qquad (G^*)^2 = -1/\lambda$$

Since $\phi''(G^*) = -2\lambda \neq 0$: **square-root branch point**. ✓

15.6 Upper Critical Dimension and the Ward Identity

The cubic vertex $\tilde{\phi}\phi^2$ has $m+n=3$, giving the naive $d_c = 4$. However, pair annihilation has a Ward identity[23] from probability conservation: $Q(z{=}0) = 0$ [30, 3]. This causes the cubic vertex coupling to be cancelled at the level of the physical process. All vertices have $n_{\text{in}} = 2$ (pure 2-particle annihilation). The physical upper critical dimension is [30]:

$$d_c^{\text{phys}} = \frac{2}{k-1} = \frac{2}{2-1} = 2 \tag{53}$$

15.7 Scaling Exponent

For pair annihilation, the density decays as $\rho(t) \sim t^{-d/2}$ for $d < 2$ and $\rho(t) \sim t^{-1}$ for $d > 2$. This is exact to all orders due to the Ward identity [30].

Verification. For $d=1$: $\rho \sim t^{-1/2}$, the Toussaint–Wilczek (1983) result [31]. ✓

For $d \geq 3$: $\rho \sim t^{-1}$ (mean-field, since $d > d_c = 2$). ✓

[23] A Ward identity is an exact algebraic relation between vertex functions that follows from a symmetry. Here the symmetry is probability conservation: the total probability $\sum_n P(n,t) = 1$ is preserved by the master equation. In the Doi formalism, this requires $Q(z{=}0) = 0$ (the Liouvillian annihilates the vacuum state). This forces certain vertex couplings to cancel, reducing the effective interaction and lowering d_c.

16 Comprehensive Validation Table

Table 1: Three-way comparison of BRW scaling exponents α_p in $\langle a^p \rangle \sim t^{\alpha_p}$. **RG**: field-theoretic calculation [4, 5]. **AC**: this worked example (formula (36)). **Sim**: Monte Carlo simulation [5, 4].

Graph	d_s	p	Theory (RG)	Theory (AC)	Simulation	
1D lattice	1	3	1/2	$(3 \cdot 1 - 2)/2 = 1/2$	0.48(4)	✓
		4	1	$(4 \cdot 1 - 2)/2 = 1$	1.0(1)	✓
		5	3/2	$(5 \cdot 1 - 2)/2 = 3/2$	1.5(1)	✓
2D lattice	2	2	1	$(2 \cdot 2 - 2)/2 = 1$	0.98(3)	✓
		3	2	$(3 \cdot 2 - 2)/2 = 2$	2.0(1)	✓
3D lattice	3	1	1/2	$(1 \cdot 3 - 2)/2 = 1/2$	0.47(2)	✓
		2	2	$(2 \cdot 3 - 2)/2 = 2$	2.0(1)	✓
5D lattice (mf)	5	1	1	$2(1) - 1 = 1$	1.0(2)	✓
		2	3	$2(2) - 1 = 3$	2.9(3)	✓
Sierpinski ($d_s \approx 1.86$)	1.86	2	0.86	$(2 \times 1.86 - 2)/2 = 0.86$	0.81(5)	✓
		3	1.79	$(3 \times 1.86 - 2)/2 = 1.79$	1.71(7)	✓
		4	2.72	$(4 \times 1.86 - 2)/2 = 2.72$	2.62(10)	✓
Random tree ($d_s = 4/3$)	4/3	2	1/3	$(8/3 - 2)/2 = 1/3$	0.35(7)	✓
		3	1	$(4 - 2)/2 = 1$	0.95(8)	✓
		4	5/3	$(16/3 - 2)/2 = 5/3$	1.6(1)	✓
Scale-free ($d_s \geq 4$, mf)	≥ 4	2	3	$2(2) - 1 = 3$	2.8(2)	✓

All 18 entries agree with simulation within statistical error.

17 What "No Loop Integral" Means

The scenic route (Feynman diagrams → RG) requires:

1. Evaluating the one-loop parametric integral $I(G; d) = \Omega_d \cdot \Gamma(-\sigma_d) \cdot \int \cdots$
2. Extracting the $1/\epsilon$ pole ($\epsilon = d_c - d$)
3. Computing the β-function
4. Solving for the Wilson–Fisher fixed point
5. Reading off the anomalous dimension

The AC route replaces *all five steps* with:

1. Build $\phi(G)$ from the vertex dictionary (algebra)
2. Find the branch point (solve a polynomial)
3. Apply the transfer theorem (proven, Cauchy integral formula)
4. Invoke $P_{\text{surv}} \sim t^{-1}$ (proven, Kolmogorov 1938)
5. Multiply by $V(t)^p$ from Weyl's law (proven, spectral geometry)

The two routes give the same answer because they detect the same singularity of the same generating function. The AC route detects it directly from the algebraic structure; the RG route detects it via the divergence structure of the loop integrals.

Remark (What IS an integral in the AC route)**.** The heat kernel Laplace transform $P_{\text{return}}(t) = \int \rho(\lambda)\, e^{-\lambda t}\, \mathrm{d}\lambda$ is an integral, but it is a *standard Laplace transform of a power law*, not a Feynman diagram loop integral. It depends on the spectral dimension d_s (a property of the graph), not on the diagram topology. The claim is not "no integrals of any kind" but rather "no Feynman diagram momentum integrals": the AC route never writes down or evaluates $\int \mathrm{d}^d k\, f(k^2)$.

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A Analytic Combinatorics in 10 Minutes

This appendix is a self-contained tutorial on the AC tools used in the main text. No prior knowledge of AC is assumed.

A.1 What is a generating function and why should you care?

Suppose you have a sequence of numbers: $a_0, a_1, a_2, \ldots$ (for instance, a_n might count the number of trees with n nodes). A **generating function** packages the entire sequence into a single function:

$$A(z) = \sum_{n=0}^{\infty} a_n\, z^n = a_0 + a_1 z + a_2 z^2 + a_3 z^3 + \cdots$$

Why bother? Because operations on the function correspond to operations on the sequence. For example:

- Multiplying two GFs *convolves* their sequences: $[z^n](A(z) \cdot B(z)) = \sum_{k=0}^{n} a_k\, b_{n-k}$.
- Composing GFs corresponds to substitution of combinatorial objects.
- Most importantly: the *singularities* of $A(z)$ in the complex plane determine the *asymptotic growth* of a_n. This is the core of AC.

The notation $[z^n]\, A(z) = a_n$ means "extract the coefficient of z^n from $A(z)$." Think of it as "reading off the n-th entry" from the encoded sequence.

Example A.1 (Geometric series)**.** The sequence $a_n = 1$ for all n gives $A(z) = \sum z^n = 1/(1-z)$. This has a **simple pole** at $z = 1$: the function blows up to $+\infty$ as $z \to 1$. The coefficients are all equal to 1 — they don't grow or decay.

Example A.2 (Catalan numbers)**.** The Catalan numbers $C_n = \frac{1}{n+1}\binom{2n}{n}$ count binary trees with n internal nodes. Their GF satisfies the self-referential equation:

$$C(z) = 1 + z\, C(z)^2$$

(a binary tree is either empty, or a root with two subtrees, each of which is itself a binary tree). Solving:

$$C(z) = \frac{1 - \sqrt{1 - 4z}}{2z}$$

This has a **square-root branch point** at $z = 1/4$: the function doesn't blow up, but the $\sqrt{}$ introduces a point where the function splits into two branches.

A.2 What is a singularity?

A **singularity** of a function $A(z)$ is a point z^* where the function ceases to be "well-behaved" (in the mathematical sense: ceases to be analytic, i.e. representable by a convergent power series).

Everyday analogy. Think of $A(z)$ as a road. A singularity is a point where the road ends — you cannot continue driving past it. Different types of singularity correspond to different ways the road ends:

- **Pole** ($1/(1-z)$): the road goes to a cliff edge and drops off to infinity. The function literally blows up.
- **Branch point** ($\sqrt{1-z}$): the road doesn't end, but it *forks*. Past the singularity, there are two roads (the + and − square roots), and you must choose which one to follow. The function is finite at the fork but its derivative blows up.

- **Logarithmic** ($\log(1-z)$): the road spirals slowly upward, never quite reaching infinity but never levelling off either.

Why do singularities matter for counting? Because the power series $A(z) = \sum a_n z^n$ converges inside a disk of radius $R = |z^*|$ (the distance to the nearest singularity) and diverges outside. The *rate* at which the coefficients grow is controlled by *how* the function fails at the singularity — gentle failure (branch point) means slow growth; violent failure (pole) means fast growth.

A.3 Why singularities determine asymptotics: the complex analysis viewpoint

The connection between singularities and coefficient growth is made rigorous by **complex analysis** — the theory of functions of complex variables $z = x + iy$.

Why complex numbers? Even if you only care about real coefficients a_n, the generating function $A(z)$ "lives" in the complex plane. A singularity at $z^* = -1/4$ (a negative real number) affects the growth of a_n just as much as one at $z^* = 1/4$. You cannot see this by looking only at real z; you need the full complex picture.

The key formula is the **Cauchy coefficient formula**:

$$a_n = [z^n]\, A(z) = \frac{1}{2\pi i} \oint_{|z|=r} A(z)\, z^{-n-1}\, \mathrm{d}z \qquad (r < R)$$

This says: to extract a_n, integrate $A(z) \cdot z^{-n-1}$ around a circle of radius r in the complex plane. (The $2\pi i$ and the contour are standard complex analysis; if you haven't seen this before, just accept it as a machine that extracts coefficients.)

The punchline. As $n \to \infty$, the factor z^{-n-1} becomes exponentially large near $|z| = R$ (the singularity) and exponentially small elsewhere. So the integral is dominated by a tiny neighbourhood of the singularity. The *shape* of the singularity (pole, branch point, log) determines the *value* of the integral, hence the growth of a_n.

This is why we care about classifying singularities: the classification directly gives us the asymptotics.

A.4 The transfer theorem: singularity type $\to$ asymptotics

The transfer theorem [6] makes this precise. The three most common cases:

Singularity type	**Local form**	$[z^n]$ **asymptotics**	**Example**
Simple pole	$\dfrac{K}{1-z/z^*}$	$K \cdot (z^*)^{-n}$	Geometric
Square-root branch	$K\sqrt{1-z/z^*}$	$\dfrac{-K}{2\sqrt{\pi}}\, n^{-3/2}\, (z^*)^{-n}$	Catalan, trees
Logarithmic	$K \log(1-z/z^*)$	$K\, n^{-1}\, (z^*)^{-n}$	Cycles

The key point: the singularity type determines the *subexponential* factor ($n^{-3/2}$, n^{-1}, constant). The exponential growth $(z^*)^{-n}$ is always determined by the *location* z^*.

A.5 Worked example: Catalan asymptotics

The Catalan GF has a square-root at $z^* = 1/4$ with local expansion:

$$C(z) \sim \frac{1}{2 \cdot 1/4} - \frac{1}{2 \cdot (1/4)} \sqrt{1-4z} = 2 - 2\sqrt{1-4z}$$

Applying the transfer theorem with $\alpha = 1/2$, $K = -2$, $z^* = 1/4$:

$$C_n = [z^n]\, C(z) \sim \frac{-(-2)}{\Gamma(-1/2)}\, n^{-3/2}\, 4^n = \frac{2}{2\sqrt{\pi}}\, n^{-3/2}\, 4^n = \frac{4^n}{\sqrt{\pi}\, n^{3/2}}$$

Verification. Stirling's approximation on $C_n = \binom{2n}{n}/(n+1)$ gives $C_n \sim 4^n/(\sqrt{\pi}\, n^{3/2})$. ✓

This is the mechanism that drives the entire paper. The Gribov DSE generates a square-root branch point just like the Catalan equation, so its coefficients also grow as $n^{-3/2}$.

B Lagrange Equations: A Tutorial

B.1 What is a Lagrange equation?

A **Lagrange equation** is a functional equation of the form:

$$T(z) = z \cdot \phi(T(z)) \tag{54}$$

where ϕ is a known function (the "kernel") and $T(z)$ is the unknown generating function. The equation says: "T is defined recursively in terms of itself."

Why do these appear everywhere in combinatorics? Because recursive structures — trees, nested parentheses, self-similar fractals — naturally produce equations where the whole is defined in terms of smaller copies of itself. A binary tree is "a root connected to two smaller binary trees"; this translates directly to $C(z) = z(1+C(z))^2$. Any time you have a combinatorial object built by gluing together smaller copies of itself, you get a Lagrange equation.

Why do these appear in field theory? Because the dressed propagator G is defined self-referentially: G is the bare propagator dressed by insertions that themselves contain dressed propagators G. This is the Dyson–Schwinger equation, and it is a Lagrange equation.

Example B.1 (Rooted labelled trees)**.** The number of rooted labelled trees on n vertices satisfies $t_n = n^{n-1}$ (Cayley's formula). The EGF $T(z) = \sum t_n z^n/n!$ satisfies:

$$T(z) = z\, e^{T(z)}$$

This is a Lagrange equation with $\phi(T) = e^T$. The e^T comes from the fact that a rooted tree is a root connected to a *set* of subtrees (and the exponential $e^x = \sum x^k/k!$ is the generating function for sets).

Example B.2 (Binary trees = Catalan)**.** The OGF for binary trees satisfies $C(z) = z(1 + C(z))^2 = z\,\phi(C)$ with $\phi(T) = (1+T)^2$. The $(1+T)^2$ comes from: each of the two children is either empty (1) or a nonempty subtree (T).

B.2 The Lagrange inversion formula

The remarkable fact about Lagrange equations is that you can compute the coefficients of $T(z)$ *without ever solving for T explicitly.* This is the Lagrange inversion formula:

The idea. If you know ϕ, you can expand $\phi(T)^n$ as a power series in T using the binomial theorem (or Taylor expansion). The n-th coefficient of $T(z)$ is then determined by a single coefficient of $\phi(T)^n$. No equation-solving required — only series expansion.

Theorem B.3 (Lagrange inversion [6, Thm. A.2])**.** *If $T(z) = z\,\phi(T(z))$ with $\phi(0) \neq 0$, then:*[24]

$$[z^n]\, T(z) = \frac{1}{n}\, [T^{n-1}]\, \phi(T)^n \tag{55}$$

[24] The formula follows from the residue theorem applied to $[z^n]T = \frac{1}{2\pi i}\oint T(z)\, z^{-n-1}\, \mathrm{d}z$. Change variables from z to T using $z = T/\phi(T)$, so $\mathrm{d}z = [\phi(T) - T\phi'(T)]/\phi(T)^2\, \mathrm{d}T$. The Jacobian and the z^{-n-1} factor combine to give $\frac{1}{n}[T^{n-1}]\phi(T)^n$. The beauty is that this extracts coefficients of $T(z)$ without ever solving for T explicitly — you only need to expand powers of ϕ.

Example B.4 (Catalan via Lagrange)**.** $\phi(T) = (1+T)^2$, so $\phi(T)^n = (1+T)^{2n}$. The coefficient of T^{n-1} in $(1+T)^{2n}$ is $\binom{2n}{n-1}$. Therefore $C_n = \frac{1}{n}\binom{2n}{n-1} = \frac{1}{n+1}\binom{2n}{n}$. ✓

B.3 Why the DSE is a Lagrange equation

Compare the Lagrange equation for binary trees with the Gribov DSE:

	Binary trees	**Gribov DSE**
Equation	$C = z(1+C)^2$	$G = G_0(1 + (\beta-\chi)G - \chi G^2)$
Variable	z = size marker	G_0 = bare propagator
Unknown	$C(z)$ = tree GF	$G(G_0)$ = dressed propagator
Kernel	$\phi = (1+T)^2$	$\phi = 1 + (\beta-\chi)G - \chi G^2$
Recursion	Root + two subtrees	Bare prop + self-energy insertions

The structures are identical: both define the unknown as a "base unit" (z or G_0) times a kernel that depends on the unknown itself. The tree equation says "a tree is a root connected to subtrees." The DSE says "a dressed propagator is a bare propagator modified by insertions that are themselves dressed."

Therefore, all the AC machinery for Lagrange equations — singularity analysis, transfer theorem, Lagrange inversion — applies directly to the DSE. This is the core insight of the paper.

B.4 Branch points of Lagrange equations

Every Lagrange equation $T = z\,\phi(T)$ with polynomial ϕ defines T as an algebraic function of z. For small z, the equation has a unique small solution (the physical one). But as z increases, a second solution approaches. At the **branch point** $z = z^*$, the two solutions merge and become indistinguishable.

Analogy. Think of the equation $x^2 = a$. For $a > 0$, there are two solutions $x = \pm\sqrt{a}$. At $a = 0$, they merge to $x = 0$. For $a < 0$, the solutions become complex. The point $a = 0$ is the branch point, and the local behaviour is $x \sim \sqrt{a}$ (a square root).

For the Lagrange equation, the same thing happens: near the branch point, $T \sim T^* + C\sqrt{z^* - z}$ (a square root). This is why almost all Lagrange equations with polynomial kernels produce square-root branch points and hence $n^{-3/2}$ coefficient asymptotics.

Recipe: solve the system $T^*\phi'(T^*) = \phi(T^*)$ and $z^* = T^*/\phi(T^*)$. If $\phi''(T^*) \neq 0$, it is a square-root branch point and the transfer theorem gives $[z^n]T \sim n^{-3/2}$. The condition $\phi''(T^*) \neq 0$ is the "generic" case; it fails only at special parameter values where the singularity upgrades to a cube root ($n^{-4/3}$) or degenerates to a pole.

This is exactly what we do in Section 6 for the Gribov DSE.

C The SIR Epidemic: A Complete AC Example on Familiar Ground

Before applying AC to the BRW, it is helpful to see the method on a system everyone has heard of: an epidemic. The SIR model is the simplest epidemic model, and its AC analysis is a direct analogue of the BRW analysis — same machinery, same singularity type, same $n^{-3/2}$ exponent. If you understand the SIR example, you understand the BRW.

C.1 The model

In the SIR model, each infected individual transmits the disease to a Poisson-distributed number of others (with mean R_0, the "basic reproduction number"), then recovers permanently. The question is: *how large is the epidemic?* — i.e. what is the distribution of the total number n of people ever infected?

This is a branching process: each infected person "branches" into R_0 new infections on average. The total size n is the total progeny of the branching tree. Its generating function satisfies:

$$T(z) = z\, e^{R_0(T(z)-1)} \tag{56}$$

This is a Lagrange equation with $\phi(T) = e^{R_0(T-1)}$.

C.2 Branch point

$\phi'(T) = R_0\, e^{R_0(T-1)}$, so:

$$T^*\, \phi'(T^*) = \phi(T^*) \quad \Longrightarrow \quad R_0\, T^* = 1 \quad \Longrightarrow \quad T^* = 1/R_0$$

$$z^* = \frac{T^*}{\phi(T^*)} = \frac{1/R_0}{e^{R_0(1/R_0-1)}} = \frac{e^{1-1/R_0}}{R_0}$$

$\phi''(T^*) = R_0^2\, e^{R_0(1/R_0-1)} = R_0^2\, e^{1-1/R_0} \neq 0$ for $R_0 > 0$. So this is a **square-root branch point**.

C.3 Coefficient asymptotics

By the transfer theorem:

$$\Pr[\text{epidemic size} = n] = [z^n]\, T \sim C\, n^{-3/2}\, (z^*)^{-n}$$

At the critical threshold $R_0 = 1$: $z^* = 1/e \cdot e = 1$ and the epidemic size distribution is $\Pr[n] \sim C\, n^{-3/2}$. This is the **Borel distribution**, a classical result in epidemiology [6].

C.4 Comparison with the BRW

The SIR and BRW analyses are structurally identical:

	SIR epidemic	**BRW (Gribov)**
Kernel	$\phi(T) = e^{R_0(T-1)}$	$\phi(G) = 1 + (\beta - \chi)G - \chi G^2$
Branch point	$T^* = 1/R_0$	$(G^*)^2 = -1/\chi$
$\phi''(T^*) \neq 0$?	Yes ($R_0^2 e^{1-1/R_0}$)	Yes (-2χ)
Singularity type	Square-root	Square-root
$[z^n]$ asymptotics	$n^{-3/2}$	$n^{-3/2}$

The $n^{-3/2}$ exponent is universal across all processes whose DSE has a quadratic kernel with $\phi''(T^*) \neq 0$. This is the directed percolation universality class.

D An Intuitive Guide to Branch Points

This appendix is for the reader who wants to *see* why branch points control the asymptotics. No formulas are needed for the core intuition — only pictures and analogies.

D.1 Three functions, three personalities

Consider three functions of a real variable x, each defined for $0 \le x < 1$:

Function	Behaviour at $x = 1$	Type	$[x^n]$ growth
$\dfrac{1}{1-x}$	Blows up to $+\infty$	Simple pole	$a_n = 1$ (constant)
$-\log(1-x)$	Grows to $+\infty$ (slowly)	Logarithm	$a_n = 1/n$ (slow decay)
$1-\sqrt{1-x}$	Finite, but slope $\to \infty$	Square-root branch	$a_n \sim n^{-3/2}$ (fast decay)

All three have a singularity at $x = 1$ (none can be extended analytically past $x = 1$). But the singularities are qualitatively different, and this difference is *exactly* what determines how the Taylor coefficients a_n behave:

- **The pole** $1/(1-x)$: the function literally blows up. It has "infinite energy" at $x = 1$. The coefficients don't decay at all — the series $1 + x + x^2 + \cdots$ has constant terms. The singularity is **violent**.
- **The logarithm** $-\log(1-x)$: the function grows to infinity, but slowly (logarithmically). The coefficients decay as $1/n$ — slow, but they do decay. The singularity is **moderate**.
- **The square root** $1-\sqrt{1-x}$: the function doesn't even blow up! It approaches a finite value ($= 1$). But its *derivative* blows up (the slope becomes vertical). The coefficients decay as $n^{-3/2}$ — much faster. The singularity is **gentle**.

Remark (The punchline)**.** The gentler the singularity, the faster the coefficients decay. A square-root branch point is the gentlest common singularity type, and it produces $n^{-3/2}$ decay. This is the exponent that appears throughout this paper.

D.2 What IS a branch point?

Consider $f(x) = \sqrt{1-x}$. For $x < 1$, this is a perfectly well-defined positive number. But what happens at $x = 1$? The function equals zero — and just past $x = 1$, say at $x = 1.01$, we would need $\sqrt{-0.01}$, which is imaginary. The function "hits a wall."

But there's more: the function $\sqrt{1-x}$ actually has *two* values for every x: $+\sqrt{1-x}$ and $-\sqrt{1-x}$. For $x < 1$, these are two distinct real numbers (one positive, one negative). At $x = 1$, they both equal zero — **they merge**. Just past $x = 1$, they become $\pm i\sqrt{x-1}$, a conjugate pair of complex numbers.

This merging point is the **branch point**. It is where two "branches" of the function (the $+$ and $-$ square roots) become indistinguishable. If you walked along the real line from left to right, you would see the two branches approaching each other, kissing at $x = 1$, and then moving apart into the complex plane.

D.3 Why branching processes have branch points

This is not a coincidence of terminology. A **branching** process (where particles split into two) naturally generates a **branch point** (where two solution sheets merge).

Here's why: the DSE $G = G_0\,\phi(G)$ with quadratic ϕ is a **quadratic equation in** G. A quadratic equation has two solutions, related by a square root (the quadratic formula). These two solutions are the two branches of the algebraic function $G(G_0)$.

At the branch point $G_0 = G_0^*$, the discriminant of the quadratic vanishes, and the two branches merge:

- The "+" branch: the physical solution (where particles survive).
- The "−" branch: the unphysical solution (which tracks extinction).

At criticality, these two solutions become degenerate — the surviving and dying processes become indistinguishable. This is exactly the **critical point** of the branching process.

D.4 The branch point as a phase transition

The analogy with thermodynamic phase transitions is exact:

Phase transition	**Branch point**
Order parameter $m(T)$	Dressed propagator $G(G_0)$
Temperature T	Bare coupling G_0
Critical point T_c	Branch point G_0^*
$m \sim (T_c - T)^{1/2}$ (mean-field)	$G \sim (G_0^* - G_0)^{1/2}$
Two phases (ordered/disordered)	Two branches (+/−)
Phases merge at T_c	Branches merge at G_0^*
Critical exponent $\beta = 1/2$	Puiseux exponent $p/q = 1/2$

The mean-field exponent $\beta = 1/2$ in the Landau theory of phase transitions is the *same* $1/2$ as the Puiseux exponent of the DSE branch point. This is not a coincidence: the Landau free energy $F(m) = am^2 + bm^4$ produces a square-root branch $m \sim \sqrt{a}$ at the critical point $a = 0$, for exactly the same algebraic reason that the DSE $G = G_0\,(1 + gG - hG^2)$ produces $G \sim \sqrt{G_0^* - G_0}$.

D.5 Why $n^{-3/2}$ and not some other exponent?

The transfer theorem says: square-root branch $\Rightarrow$ $n^{-3/2}$ coefficients. But *why* $-3/2$ specifically?

Rough argument. The coefficients $a_n = [z^n]f(z)$ are extracted by the contour integral $a_n = \frac{1}{2\pi i}\oint f(z)\,z^{-n-1}\,\mathrm{d}z$. Near the branch point at $z = z^*$, the dominant contribution comes from a small neighbourhood of z^*. Parameterize $z = z^*(1 - t/n)$ for small t. Then:

- $f(z) \sim \sqrt{t/n}$ (from the square root).
- $z^{-n} \sim (z^*)^{-n}\,e^t$ (from the exponential).
- The integration measure contributes $\mathrm{d}z \sim z^*/n\,\mathrm{d}t$.

Putting it together:

$$a_n \sim (z^*)^{-n}\int_0^\infty \frac{\sqrt{t/n}}{n}\,e^{-t}\,\mathrm{d}t = (z^*)^{-n}\,n^{-3/2}\int_0^\infty \sqrt{t}\,e^{-t}\,\mathrm{d}t$$

The integral is $\Gamma(3/2) = \sqrt{\pi}/2$ (a constant). So $a_n \sim C\,n^{-3/2}\,(z^*)^{-n}$.

The $-3/2$ comes from two powers of n in the denominator: one from the square root ($n^{-1/2}$) and one from the integration measure (n^{-1}). A simple pole would give n^{-1} from the measure alone (no square root), hence $a_n \sim (z^*)^{-n}$ (constant coefficients).

D.6 Visualising the Gribov DSE

The Gribov DSE $G = G_0(1 + (\beta{-}\chi)G - \chi G^2)$ defines G as a function of G_0. For real values of the couplings:

- For small G_0 (weak coupling): there is one real solution G, close to G_0 (the perturbative regime). The Taylor series $G = G_0 + c_2 G_0^2 + c_3 G_0^3 + \cdots$ converges.
- As G_0 increases toward G_0^*: the solution curve bends, the derivative $\mathrm{d}G/\mathrm{d}G_0$ grows, and a second (unphysical) solution approaches from the other direction.
- At $G_0 = G_0^*$ (the branch point): the two solutions merge. The derivative $\mathrm{d}G/\mathrm{d}G_0 \to \infty$ (vertical tangent). The Taylor series has radius of convergence $|G_0^*|$.
- For $G_0 > G_0^*$: the two solutions become complex conjugates. The Taylor series diverges. This is the strong-coupling (non-perturbative) regime.

The branch point is the *boundary* between the perturbative and non-perturbative regimes. The transfer theorem extracts the asymptotic behaviour of the Taylor coefficients from the *shape* of this boundary — specifically, from whether it is a square root (gentle) or something sharper.

Remark (The one sentence summary)**.** The branch point of the DSE is the critical point of the theory. Its type (square-root, cube-root, pole) is the universality class. The transfer theorem converts the type into a number ($n^{-3/2}$, $n^{-4/3}$, constant). That number determines all the critical exponents. That is the entire AC method.

E Notation Reference

For the reader's convenience:

Symbol	**Meaning**
A	Particle species (active walker)
β, ϵ, χ	Branching, death, coagulation rates
S	Stoichiometry matrix (rows = reactions, columns = (k, ℓ))
z, ∂_z	Heisenberg–Weyl generators (z = creation, ∂_z = annihilation)
Q	Liouvillian (infinitesimal generator of the master equation)
$\tilde{\phi}, \phi$	Response field and field in the Doi–Peliti action
g_{mn}	Coupling constant of vertex $\tilde{\phi}^m \phi^n$
G, G_0	Dressed and bare propagator
$\phi(G)$	DSE kernel (Lagrange equation: $G = G_0\, \phi(G)$)
G^*, G_0^*	Branch point location
$[z^n]A(z)$	Coefficient of z^n in the power series $A(z)$
$P_{\text{surv}}(t)$	Survival probability of the BRW at time t
$P_{\text{return}}(t)$	Return probability of a random walk at time t
$V(t)$	Volume (number of distinct sites) explored by time t
d, d_s, d_c	Euclidean, spectral, and upper critical dimension
α_p	Scaling exponent: $\langle a^p \rangle \sim t^{\alpha_p}$
$\lvert\mathrm{Aut}(\Gamma)\rvert$	Size of the automorphism group of diagram Γ